

Stacked-based machine learning to predict the uniaxial compressive strength of concrete materials

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ABSTRACT

Compressive strength is a key factor in the design and durability of concrete structures. Accurate prediction of compressive strength helps optimize material use and reduce construction costs. This study proposes a novel stacked model for predicting compressive strength, integrating three base models with linear regression. The base models include Artificial Neural Networks, Random Forest, and Extreme Gradient Boosting, while the stacked model uses Linear Regression as the metamodel. A dataset of 1,030 concrete mix samples covering eight critical input parameters, including cement, blast furnace slag, coarse aggregates, fine aggregates, fly ash, water, superplasticizer, and curing days, was used for training and evaluation. The dataset was split into training (80%), validation (10%), and testing (10%) subsets. The models were trained independently, and their predictions were used to develop the stacked model. Among the base models, the Extreme Gradient Boosting model achieved the highest accuracy, with an R^2 of 0.947 during testing. However, the stacked model outperformed it, attaining an R^2 of 0.953 in the testing phase. Shapley additive explanations analysis identified curing duration as the most influential factor in compressive strength prediction. A user-friendly graphical interface was developed to facilitate efficient prediction of compressive strength in concrete structures.

1. Introduction

The increasing release of greenhouse-gases like carbon dioxide, methane, and nitrous oxide into the atmosphere, primarily driven by human activities such as burning fossil fuels, deforestation, and industrial processes, is causing profound changes to the Earth's environment. This rise leads to widespread climate disruptions, including shifts in weather patterns, more frequent extreme weather events, and severe impacts on ecosystems [1]. A particularly visible consequence is the reduction of polar ice stores in the Arctic and Antarctic regions, where ice caps and glaciers are melting at an accelerating rate. This reduction poses significant ecological challenges, including rising sea levels and disrupted ecosystems, which threaten biodiversity and the livelihoods of millions of people worldwide. Among the contributors to these emissions, the building industry, a key sector within the global economy, emerges as the largest contributor to greenhouse-gas emissions, accounting for nearly 50 % of the total worldwide emissions. This substantial impact underscores the industry's pivotal role in global environmental concerns [2]. This is primarily due to the extensive use of concrete, which relies heavily on Portland cement (PC) as a key

component [3]. PC is a major driver of CO₂ emissions, accounting for approximately 7 % of the total global emissions [4]. The production process of PC involves the calcination of limestone (calcium carbonate), which releases carbon dioxide. Notably, the calcination process, where calcium oxide (CaO) is produced, is responsible for a significant portion of these emissions. As urbanization and infrastructure development continue to expand, the demand for concrete and consequently PC has surged. Presently, the annual production of PCs stands at 4,000 million tons, with forecasts predicting this figure will rise to 6,000 million tons by 2060 [5]. This trend exacerbates environmental concerns and highlights the need for sustainable alternatives. Addressing this critical issue requires the development of effective substitutes that can satisfy the increasing demand while drastically lowering energy consumption and CO₂ emissions [6]. One promising approach is incorporating waste and recycled materials into concrete production. This method not only addresses the demand for construction materials but also contributes to reducing environmental hazards and conserving natural resources. By diverting industrial waste from landfills and utilizing it in concrete, we can mitigate the ecological footprint of construction activities [7,8].

Various industrial by-products have been recognized as suitable alternatives to PC in concrete production. Some of the most promising

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Nomenclature

AI	Artificial Intelligence
ANFIS	Adaptive Neuro-Fuzzy Inference System
ANN	Artificial Neural Network
AdaBoost	Adaptive Boosting
BP	Back Propagation
BM s	Base Models
BO	Bayesian Optimization
BiLSTM	Bidirectional Long Short-Term Memory
C	Cement
CC	Correlation Coefficient
CNN	Convolutional neural network
CPI	Cumulative Prediction Error Index
CS	Compressive Strength
CV	Cross-validation
CaO	Calcium Oxide
Cagg	Coarse aggregate
cov	Covariance
DL	Deep Learning
DT	Decision Tree
FA	Fly Ash
Fagg	Fine aggregate
GA	Genetic Algorithm
GB	Gradient Boosting
GBM	Gradient Boosted Machine
GBR	Gradient Boosting Regressor
GBRT	Gradient Boosted Regression Tree
GEP	GeneExpression Programming
GRU	Gated Recurrent Units
GUI	Graphical User Interface
HPC	High Performance Concrete
HSC	High Strength Concrete
IQR	Interquartile Range
KDE	Kernel Density Estimation

LR	Linear Regression
LSC	Lightweight Self-compacting Concrete
LSTM	Long Short-Term Memory
LWC	Lightweight Concrete
LightGBM	Light Gradient Boosting Machine
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MBE	Mean Bias Error
ML	Machine Learning
MLPNN	Multilayer Perceptron Neural Network
MSSA-DE	Multi-objective Metaheuristic Differential Evolution
OPC	Ordinary Portland Cement
PC	Portland Cement
PDP	Partial Dependence Plot
PDPs-1D	One-Dimensional Partial Dependence Plot
RAC	Recycled Aggregate Concrete
RF	Random Forest
RMSD	Root Mean Square Difference
RMSE	Root Mean Square Error
RNN	Recurrent Neural Networks
R²	Determination coefficient
r	Pearson correlation coefficient
SCC	Self-Compacting Concrete
SD	Standard Deviation
SHAP	Shapley Additive exPlanations
SM	Stacked Model
SP	Superplasticizer
SVM	Support Vector Machines
TR	Training stage
TS	Testing stage
UHPC	Ultra-High Performance Concrete
VAF	Variance Accounted For
VD	Validation stage
W	Water
XGB	Extreme Gradient Boosting

options include ground granulated blast-furnace slag [9], granite powder [10], and fly ash [11]. These materials are frequently produced in significant amounts during industrial activities and often present challenges related to disposal. By incorporating these materials into concrete, we can not only improve the properties of the concrete, such as its durability and strength but also significantly reduce its environmental impact. For instance, replacing traditional PCs with alternative materials like ground granulated blast-furnace slag, granite powder, and fly ash can significantly reduce greenhouse-gas emissions, potentially reducing up to 80 % [12]. This significant reduction highlights the potential of these materials to contribute to more sustainable construction practices [13].

Concrete remains the most widely used construction material globally, surpassed only by water consumption [14–17]. Its widespread availability, malleability upon cooling, and strength to support loads after hardening are reasons for its widespread acclaim. These properties make concrete an essential material for various construction applications. The performance of concrete, particularly its compressive strength, is critical for the safety and durability of structures [18]. Compressive strength (CS) refers to the material's ability to withstand loads that compress or reduce its size. This property is influenced by various factors, including the water-to-binder ratio, the size and quality of aggregates, and the type of binder used [19]. The heterogeneous nature of concrete, which comprises a mix of binder, aggregates, water, and admixtures, adds to the complexity of predicting its compressive strength. Each component and how they are combined can significantly affect the final properties of the concrete. For example, a lower water-to-

binder ratio can lead to higher compressive strength but may reduce workability, while aggregates' size and grading can impact the finished product's density and durability. The type of binder, whether it is a traditional PC or an alternative material, also plays a crucial role in determining the concrete's performance [20].

Traditionally, the evaluation of concrete's compressive strength has depended on laboratory testing techniques, comprising crushing concrete samples with predetermined dimensions following a designated curing time. This method is universally accepted due to its ability to provide reliable measurements of a material's load-bearing capacity [21]. Despite its widespread use, this approach is increasingly viewed as inefficient. The process is both time-intensive and costly, as it requires significant labor and specialized equipment. Moreover, it involves destructive testing, which does not allow for further use of the tested samples, thereby adding to material waste and cost. To overcome the limitations of traditional testing methods, some researchers have explored using empirical regression techniques to predict the CS of concrete based on its mix proportions [22–24]. These techniques involve creating mathematical models that estimate strength outcomes from the known quantities of the mixture's ingredients, such as cement, water, aggregates, and admixtures. However, developing accurate regression models is challenging due to the intricate and nonlinear relationships among the various components of concrete. Factors such as the type and fineness of cement, the water-to-cement ratio, the size and quality of aggregates, and the presence of additives can all significantly influence the strength, making it difficult to predict outcomes precisely [25].

In recent years, advancements in artificial intelligence (AI) and

machine learning (ML) have revolutionized various industries, including construction. AI, particularly through the use of ML models such as Artificial-Neural-Networks (ANN), Support-Vector-Machines (SVM), Gene-Expression-Programming (GEP), and Deep Learning (DL), has demonstrated remarkable capabilities in handling complex data sets and extracting meaningful patterns [26–30]. These models have been applied to a wide range of problems, providing more efficient and accurate solutions compared to traditional statistical methods. Applications aimed at predicting concrete mechanical properties emphasize various methods, challenges, and their practical implications, underscoring the expanding role of AI in engineering [31,32]. Techniques have also been utilized to optimize concrete mix design, enhancing the reliability and efficiency of construction processes [33]. In structural engineering, ML has been widely used to predict the behavior of structural elements [34–38], detect damage [39,40], and calculate buckling loads [41,42]. Many studies have employed ML techniques to estimate the axial load capacity of composite columns [43–48].

Recent reviews also highlight the role of ML in predicting the behavior of concrete under extreme conditions, such as elevated temperatures [49]. Additionally, heuristic algorithms, combined with semi-empirical formulas, have been successfully employed for compressive strength estimation, offering more transparent relationships between input variables and outcomes [50]. Furthermore, hybrid ML approaches integrating ANN and particle swarm optimization (PSO) have been applied in various studies, such as predicting soil thermal conductivity and improving concrete property forecasts' accuracy [51]. Moreover, innovative techniques have been used to predict the unconfined compressive strength of granite, employing only two non-destructive testing indexes, demonstrating the potential of ANN models in diverse geological applications [52].

In the context of concrete technology, ML models can be trained on large datasets of past concrete mix compositions and their corresponding compressive strengths, allowing for the creation of predictive models that can forecast the strength of new mixtures with high accuracy. Machine learning offers several advantages over traditional empirical regression methods, particularly in handling the complexities and non-linearities inherent in concrete mixture design. ML models can learn from vast amounts of data and identify subtle patterns that might not be evident through conventional methods. This ability to model complex relationships makes ML particularly useful for predicting the CS of concrete, including variations with alternative materials like fly ash. Furthermore, ML models can adapt and improve over time as more data becomes available, continually refining their predictions.

Fly ash, a by-product of coal combustion has emerged as a valuable alternative material in concrete production due to its environmental and performance benefits. Incorporating fly ash into concrete not only helps in waste management by reducing the need for landfill space but also improves the mechanical properties of concrete, such as its durability and resistance to chemical attack. Using ML models to predict the CS of fly ash-based concrete is essential for optimizing mix designs. Accurate predictions ensure that the concrete meets necessary performance standards while also maximizing the use of sustainable materials. This approach helps in achieving a balance between structural integrity and environmental sustainability. Various factors affect the CS of concrete, such as the water-to-binder ratio, the size and quality of aggregates, the type of binder employed, the curing period, and the distinct properties of alternative materials like fly ash. The water-to-binder ratio, for instance, directly affects the porosity of the concrete, which in turn influences its strength and durability. Similarly, aggregates' size and grading can impact the mix's workability and CS. Understanding these factors and their interactions is critical for designing concrete mixtures that meet specific strength and performance criteria.

Implementing ML in analysing large datasets from past concrete mixtures allows for developing models that accurately predict concrete characteristics. By examining historical data, ML models can optimize the proportions of materials in new mixtures, enhancing quality control

and consistency in concrete production. These models are particularly effective in identifying the optimal balance of materials that achieve desired strength properties while minimizing costs and environmental impact. The use of ML in this context improves the efficiency of concrete design and supports the development of more sustainable building practices. ML models play a pivotal role in promoting the sustainability of concrete mix designs. ML can help reduce the environmental footprint of concrete production by optimising material combinations. For instance, using supplementary cementitious materials like fly ash can lower carbon emissions associated with cement production, significantly contributing to global CO₂ emissions. Additionally, optimizing the mix design can reduce material waste and improve concrete structures' overall lifespan and durability, leading to long-term sustainability benefits.

The application of ML in structural engineering has gained significant attention, particularly in predicting the concrete's CS. Various studies have demonstrated the potential of ML models in this domain. For instance, Chithra et al. [53] successfully utilized Artificial Neural Networks (ANN) to predict the CS of high-performance concrete (HPC) containing silica nanoparticles and copper slag, showcasing the ANN model's ability to handle complex material properties. Similarly, Nguyen et al. [54] developed four different ML models to forecast the compressive and tensile strength of HPC, with their findings emphasizing the superior accuracy of Gradient-Boosting-Regressor (GBR) and Extreme-Gradient-Boosting (XGB) models in capturing intricate data patterns. Kumar et al. [55], focusing on lightweight concrete (LWC), employed SVM models based on data from 120 experimental sets, highlighting SVM's effectiveness in evaluating LWC compressive strength across diverse datasets. In another study, Ashrafian [56] applied heuristic regression methods to analyze the strength and ultrasonic pulse velocity of fibrous concrete, providing valuable insights into the material's behavior under various conditions. Using a comprehensive investigation of eight input factors crucial to the model's accuracy, Zhang et al. [57] employed the Random Forest (RF) approach to forecast the uniaxial CS of lightweight self-compacting concrete (LSC).

Furthermore, the incorporation of nanotechnology, particularly Nanosilica, has also been explored to enhance the compressive strength of concrete, with various models assessing the influence of mix proportions and curing conditions on the material's strength [58]. Aslam et al. [59] developed a GEP model to predict the compressive strength of high-strength concrete (HSC), leveraging a comprehensive dataset of 357 points to illustrate GEP's robustness in modeling complex material characteristics. Collectively, these studies underscore the diverse applications and effectiveness of various ML models in predicting concrete properties, highlighting the significant advancements in this field. The overview of these ML models and approaches used in past literature is further detailed in Table 1, providing a comprehensive understanding of the advancements in this field.

2. Research significance

The novelty of this research resides in its thorough methodology for using machine learning models to predict CS and developing an innovative stacked linear regression model (SM-LR), which provides substantial enhancements in accuracy and practical usability compared to conventional techniques. This model integrated three ML models: ANN, RF, and XGB, as the base models. These models were chosen based on their strong predictive capabilities, which, combined with stacking, can effectively overcome the weaknesses of each model. These selected models align better with the dataset's size and complexity, avoiding the need for extensive data, which deep learning models require. They strike a balance between predictive performance and the ability to interpret how inputs affect outputs, which is crucial for applications in structural engineering where model insights guide decisions. Given the available computational resources, these models offer a practical compromise between accuracy and computational demands. The literature review

Table 1
Distinct ML models applied in previous literature.

Concrete type	Dataset	Model	Performance index of the best model	Ref.
HPC	471	MLPNN, DT, RF, and GBRT	$R^2 = 0.89$, error less than 10 MPa.	[60]
HPC	1030	GBRT	$R^2 = 0.92$, MSE = 22.09 MPa ² , RMSE = 4.7 MPa.	[25]
HSC	225	FA and RF	$R^2 = 0.9747$ (Training), 0.8753 (test); RMSE = 3.6 MPa (Training), 6.6 MPa (test).	[61]
HPC	1030	XGB, RF, and SVR	$R^2 = 0.95$, RMSE = 0.05, MAE = 0.15.	[62]
HPC	270	GEP, DT, and Bagging	$R^2 = 0.911$	[63]
UHPC	78	ANN	$R^2 = 0.9986$, MAPE = 0.0026 %, RMSE = 4.6564 %.	[64]
HPC	1761	SVM, ECHO-SVM, ANN, ANFIS, ELM, M5, and NLSR	$R^2 = 0.9418$, SCC = 0.9369, MAPE = 14.013, MAE = 3.9797, RRMSE = 1.0463.	[65]
SCC	7	ANFIS	$R = 95.79$ % (Average of all testing datasets).	[66]
HPC	1133	MFA-ANN	$R = 0.95$.	[67]
Pozzolanic concrete	310	GEP and ANN	R^2 (Training) = 0.94, R^2 (Testing) = 0.88.	[68]
OPC and HPC	2817	ANN and ANFIS	$R^2 = 0.9675$, RMSE = 4.6877 MPa, MAE = 3.5349 MPa, MBE = 0.3829 MPa.	[69]
HSC	357	GEP	$R^2 = 0.914$, RRMSE = 0.031, RSE = 0.023, MAE = 0.595, RMSE = 1.62.	[59]
UHPC	626	XGB, LightGBM, hybrid XGB-LightGBM	$R^2 = 0.996$ (Training), 0.963 (Testing).	[70]
RHA-FA concrete	138	ANN, XGB, GBM	$R^2 = 0.9933$ (Train), 0.8404 (Test), RMSE = 0.6755 (Train), 3.2776 (Test), MAPE = 9.3646 (Test)	[71]
UHPC	810	AB, RF, GB	$R^2 = 0.9913$ (Train), 0.9804 (Test).	[72]
Recycled aggregate concrete (RAC)	105	MSSA-DE	$R^2 = 0.9906$ (Train), 0.9604 (Test).	[73]
HPC	2171	BiLSTM, CNN, GRU, LSTM	$R^2 = 0.990$ (Training), $R^2 = 0.961$ (Testing).	[74]
HPC	1030	ANN, MARS-L, MARS-C, GPR, MPMR, and HENSIM.	$R^2 = 0.8894$, RMSE = 0.0701, MAE = 0.0490, MAPE = 16.2757, a-20 index = 78.6408.	[75]
Lightweight Structural Concrete	449	MARS, M5P, ELM, and GEP	$R^2 = 0.966$, RMSE = 4.549 MPa, RAE = 2.484 MPa.	[76]
Recycled Aggregate Concrete (RAC)	209	ICA-XGBoost, ICA-ANN, ICA-SVR, and ICA-ANFIS	$R^2 = 0.983$, MAPE = 0.031, MAE = 1.147, RMSE = 1.479, a10-index = 1.000.	[77]

confirms their successful application in similar engineering contexts, validating their effectiveness and appropriateness for the study's objectives. The following are the authors' specific contributions:

1. To introduce a stacked model (SM) development approach that integrates three base ML models: ANN, RF, and XGB, with a linear regression (LR) *meta*-model and to assess their effectiveness in predicting concrete's CS.
2. Utilizing Bayesian optimization for hyperparameter tuning significantly enhances the performance of the base models.

3. A sensitivity analysis was performed using Shapley-Additive-exPlanations (SHAP) to identify the primary variables that impact CS prediction, providing useful insights for enhancing accuracy.
4. Conducting PDP analysis to illustrate the functional relationship between specific input parameters and model predictions.
5. This work highlights the importance of blending theoretical model development with practical usability by developing an intuitive Graphical User Interface (GUI), which helps to close the gap between sophisticated computational predictions and their effective application in real-world scenarios.

3. Research objectives

This study extends prior research by utilizing ML models to predict the CS of concrete within a data range (2 to 82 MPa). The primary aim was to assess the effectiveness of various base ML models in predicting CS and then enhance their accuracy by stacking techniques. The research methodology, illustrated in Fig. 1, involved collecting and analyzing data from 1030 datasets, including parameters such as cement and aggregates, with the properties examined through histograms and heatmaps. This study utilized three ML models to predict the CS of concrete. The models' effectiveness was assessed by comparing the actual values with the predicted outcomes, employing various regression metrics. To ensure reliable results, k-fold cross-validation was used throughout the evaluation process. Additionally, an in-depth sensitivity analysis was performed to explore how different variables affect model accuracy. The findings were compared with prior studies to pinpoint the most accurate model. This methodology presents a more efficient and economically viable alternative to conventional experimental procedures, offering deeper insights into the factors influencing concrete CS prediction.

4. Methodology

4.1. Database collection

Using information gathered from previous studies or data collected in the lab, researchers are able to develop predictive models. This study conducted a review of previous academic literature, including notable works by Song et al. [19,60] and Yeh [78], to compile a dataset comprising 1,030 data points pertinent to the CS of concrete. The primary features used as input variables in the analysis included cement (C), blast furnace slag (Slag), fly ash (FA), water (W), superplasticizer (SP), coarse aggregate (Cagg), fine aggregate (Fagg), and the curing duration (Age). These factors were crucial in determining the final compressive strength outcome.

4.2. Descriptive statistics

Table 2 offers a summary of the collected data's quantitative features, detailing each of its features in detail. The rows provide detailed statistical data for each variable, while each column represents a distinct variable.

The Kernel Density Estimation (KDE) plots are shown in Fig. 2. It provides a comprehensive overview of the distribution characteristics for the dataset's nine distinct variables (X1 through X8 and Y). Key statistical indicators included in these plots are the mean, standard deviation, skewness, and kurtosis, which correspond to the quantitative data shown in Table 2. The mean provides the average value of the data, representing the central tendency. The standard deviation measures the dispersion or spread of the data around the mean. Skewness quantifies the asymmetry of the data distribution: positive skewness indicates a longer right tail, while negative skewness points to a longer left tail. Kurtosis measures the "tailedness" of the distribution: high kurtosis (>3) indicates heavy tails and peakiness, while low kurtosis (<3) suggests lighter tails and a flatter distribution.

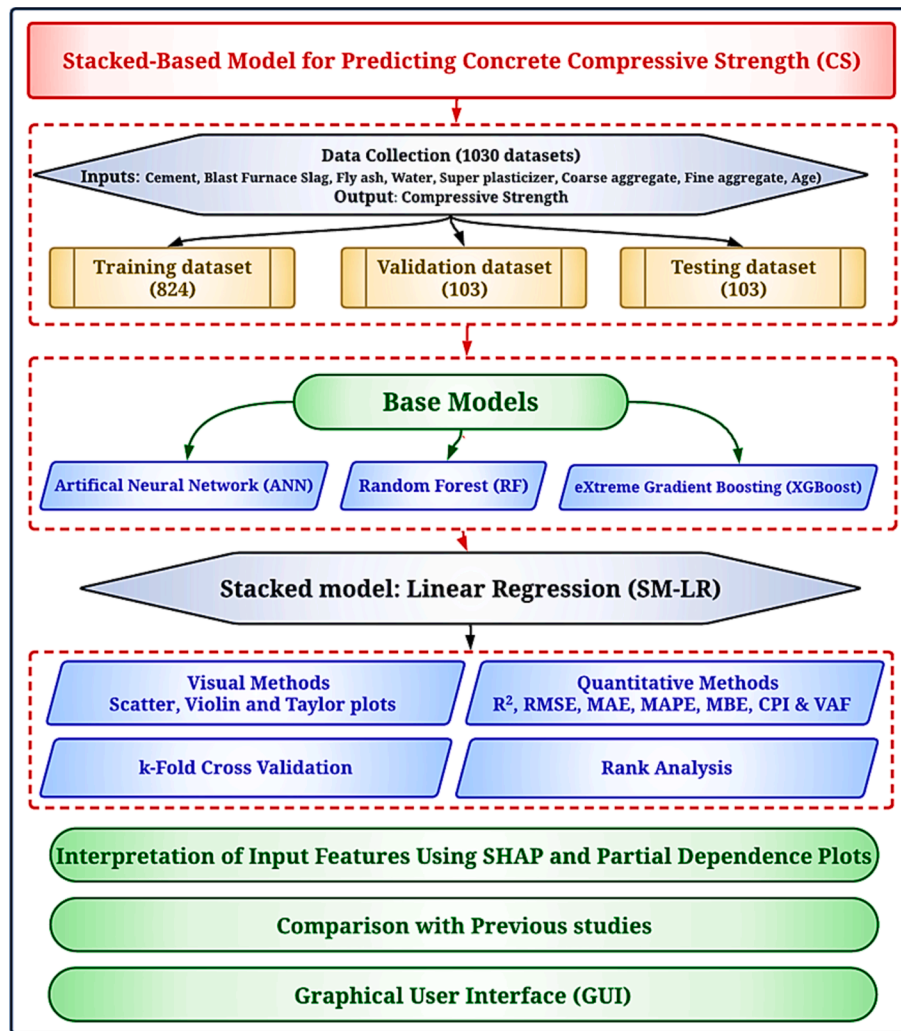


Fig. 1. Methodology flowchart.

Table 2

Summary of the statistical information for the dataset.

Variables → Statistic ↓	C (kg m ⁻³)	Slag (kg m ⁻³)	FA (kg m ⁻³)	W (kg m ⁻³)	SP (kg m ⁻³)	C _{agg} (kg m ⁻³)	F _{agg} (kg m ⁻³)	Age (days)	CS (MPa)
Mean	X1 281.17	X2 73.90	X3 54.19	X4 181.57	X5 6.20	X6 972.92	X7 773.58	X8 45.66	Y 35.82
Standard Deviation	104.5	86.3	64.0	21.4	6.0	77.8	80.2	63.2	16.7
Minimum	102.0	0.0	0.0	121.8	0.6	801.0	594.0	1.0	2.33
Median	272.9	22.0	0.0	185.0	6.4	968.0	779.5	28.0	34.4
Maximum	540.0	359.4	200.1	247.0	32.2	1145.0	992.6	365.0	82.6
Count	1030	1030	1030	1030	1030	1030	1030	1030	1030

For instance, the input **X1**, with a mean of 281.17 and a standard deviation of 104.51, shows a relatively normal distribution with slight positive skewness (0.51) and moderate kurtosis (−0.32), suggesting a concentration of data points around the mean with a few outliers. The input **X2**, having a mean of 78.90 and a standard deviation of 86.28, reveals more pronounced positive skewness (0.83) and kurtosis (−0.41), highlighting a higher frequency of lower values and the presence of significant outliers. The input **X3**, with a mean of 54.19 and a standard deviation of 64.00, demonstrates a bimodal distribution indicative of two dominant data clusters, also reflected in its negative kurtosis (−1.33). Similarly, the input **X4**'s mean of 181.57 and standard deviation of 21.86, coupled with near-zero skewness (0.17) and kurtosis (0.12), point to a distribution that is fairly symmetrical but slightly leptokurtic. The input **X5**, with a mean of 6.20 and a standard deviation

5.97, exhibits a multi-modal distribution with substantial skewness (0.91) and kurtosis (1.41), indicating complex underlying data structures. The input **X6**, with a mean of 972.92 and a standard deviation of 77.58, displays a near-normal distribution, albeit with slight positive skewness (0.44) and kurtosis (−0.60). The input **X7**'s mean of 773.58 and standard deviation of 20.18, along with a high kurtosis value (−0.10) and slight negative skewness (−0.25), suggest a sharp peak around the mean. The input **X8**, characterized by a mean of 45.66 and a standard deviation of 63.17, presents a complex distribution with high skewness (2.77) and kurtosis (7.17), indicating a high frequency of lower values and the presence of significant outliers. Lastly, the response output (**Y**), with a mean of 35.82 and a standard deviation of 16.71, exhibits a relatively normal distribution with moderate positive skewness (0.31) and kurtosis (−0.31), suggesting a well-centred dataset with

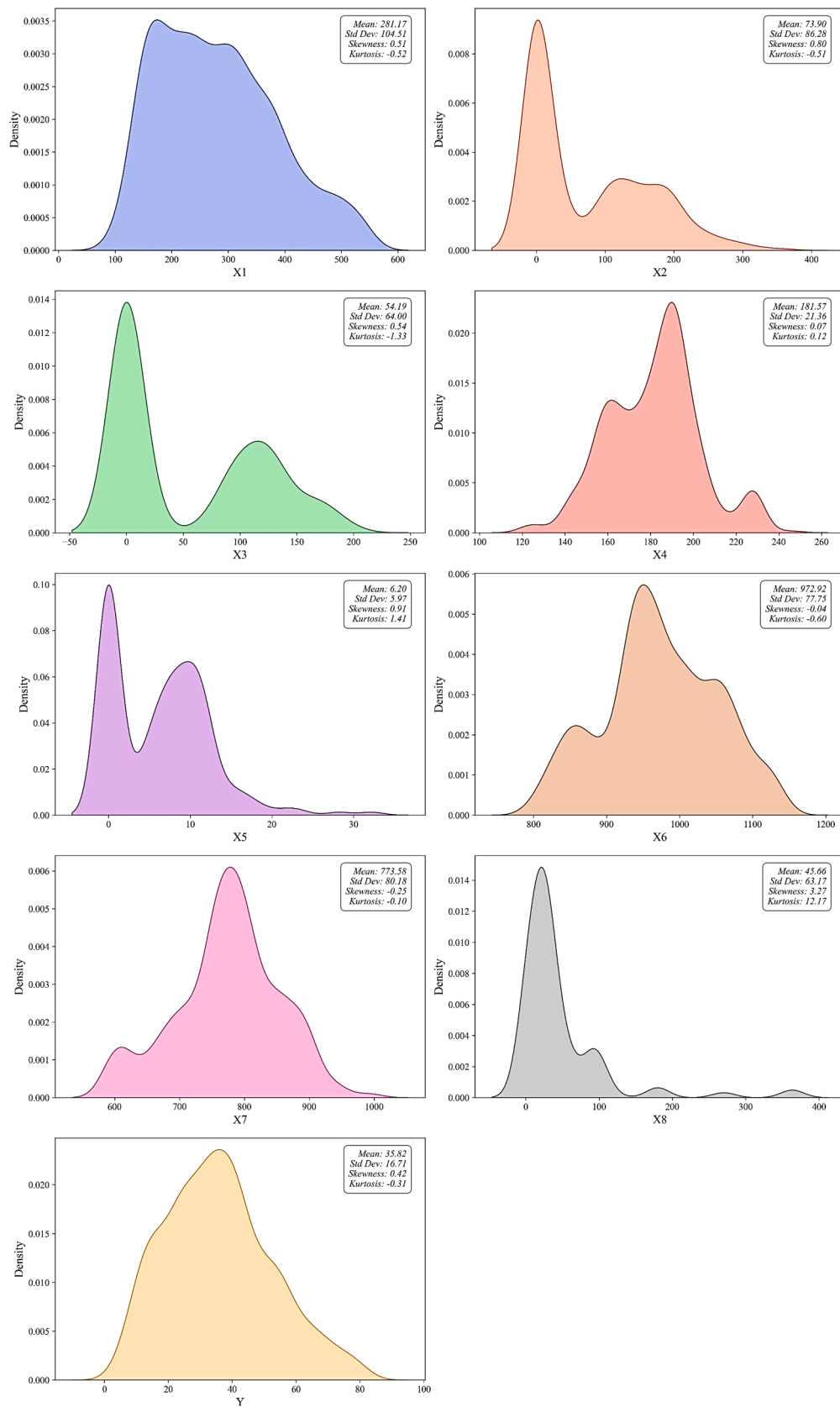


Fig. 2. KDE plots of input and output variables.

some outliers.

In summary, most of the variables (Slag, FA, SP, and Age) display a noticeable positive skew, indicating a higher concentration of lower values with fewer higher values. Conversely, the variables W, Cagg, Fagg, and CS show more evenly distributed data, with central tendencies being more apparent.

4.3. Correlation analysis

The objective of this analysis is to find and determine the best prediction model by dissecting inter-variable. It is essential to comprehend these relationships in order to fully appreciate the connections between the desired strength factor and its dependent features. The Pearson correlation coefficient (r) is the most popular metric for this approach since it provides insight into these connections [79,80]. It is defined in Eq. (1) as the ratio of the product of the standard deviations of two variables (x, y) to their covariance (cov).

$$r = \frac{cov(x, y)}{\sigma_x \sigma_y} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (1)$$

The average value of two variables, x and y , is denoted as $\bar{x}$ and $\bar{y}$, respectively, while the number of datasets is denoted by (n). A heatmap showing the interdependence of all variables is shown in Fig. 3.

According to the heatmap, the variables C, SP, and Age exhibit the strongest positive correlations, with r -values of 0.498, 0.366, and 0.329, respectively. These values indicate that these variables are the most influential. This suggested that It becomes increasingly apparent that the addition of cement, superplasticizer, and curing time significantly impacts the compressive strength of concrete as the amount of these additives increases. Given that the most pronounced negative correlation between X4 and X5 (-0.657) is found between the two variables, it would appear that water and superplasticizers have a relationship that is opposing to one another. In addition, the coefficient of correlation CS of concrete and water have a strong negative correlation, with an r -value of -0.290 . There is only a weak relationship between the remaining variables and the concrete's CS, which suggests that the variables do not

form linear solid relationships with one another. Since there are no uncorrelated features, we can conclude that all eight input parameters help predict concrete CS.

4.4. Hixbin graphs

Fig. 4 presents hexbin plots revealing important relationships between the dependent variable (CS of concrete) and the various input variables. For cement vs. CS, the hexbin plot shows a strong positive correlation. As cement content increases, the CS of concrete also increases. The data points show a high density in the middle to upper ranges of cement. The concentrated clustering around the higher cement values suggests that cement significantly and consistently impacts increasing CS. Higher cement content correlates strongly with increased CS, with the densest areas appearing in the higher range of cement content. The plot shows a moderate positive correlation for Blast Furnace Slag vs. CS. While there is a noticeable upward trend, the data points are more dispersed compared to cement. The highest concentration of data points occurs in the middle range of Slag, implying that moderate amounts of Slag contribute most significantly to CS. This suggests that while Slag positively impacts CS, its effect is most noticeable at intermediate levels.

The fly ash (FA) plot shows a weak positive correlation with CS. The data points are scattered across the plot with a lower density in the higher ranges of fly ash. The density appears to be more concentrated in the lower to middle ranges of FA, which suggests that while fly ash may improve CS slightly, its effect is less pronounced and more variable, particularly at higher levels. Meanwhile, the hexbin plot of water vs. CS reveals a negative correlation. As water content increases, CS decreases. The data points are most concentrated in the lower to middle ranges of Water, with high data density around lower water content values. This supports the idea that lower water content correlates with higher CS, whereas higher water content leads to weaker concrete, as indicated by the concentration of points in the lower ranges. The Superplasticizer (SP) versus CS plot reveals a moderate positive correlation. The data points show a concentration in the middle to upper ranges of SP. This suggests that higher amounts of SP contribute to improved CS, likely due to better workability without sacrificing strength.

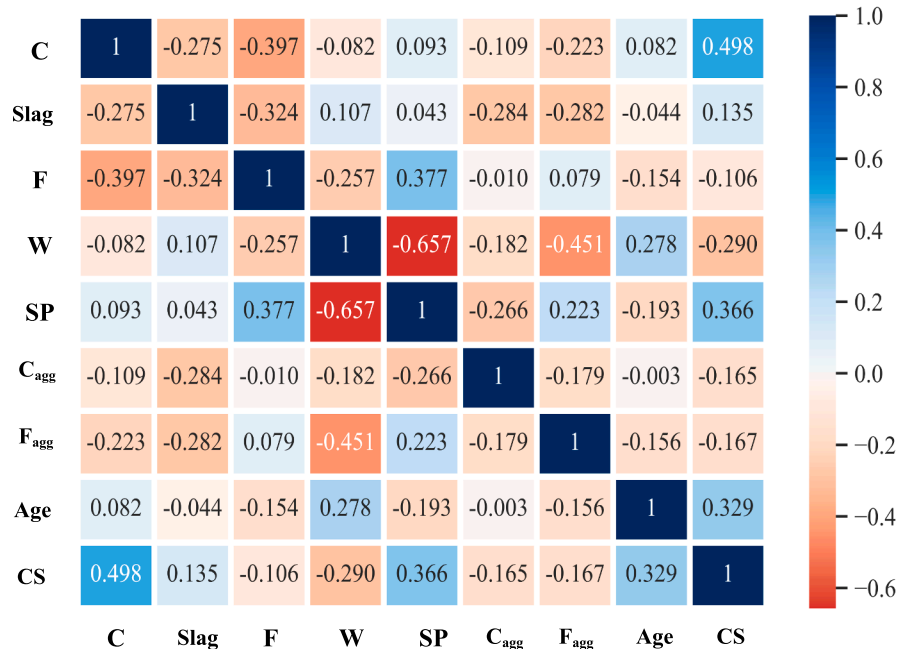


Fig. 3. Heatmap correlation matrix for all studied parameters.

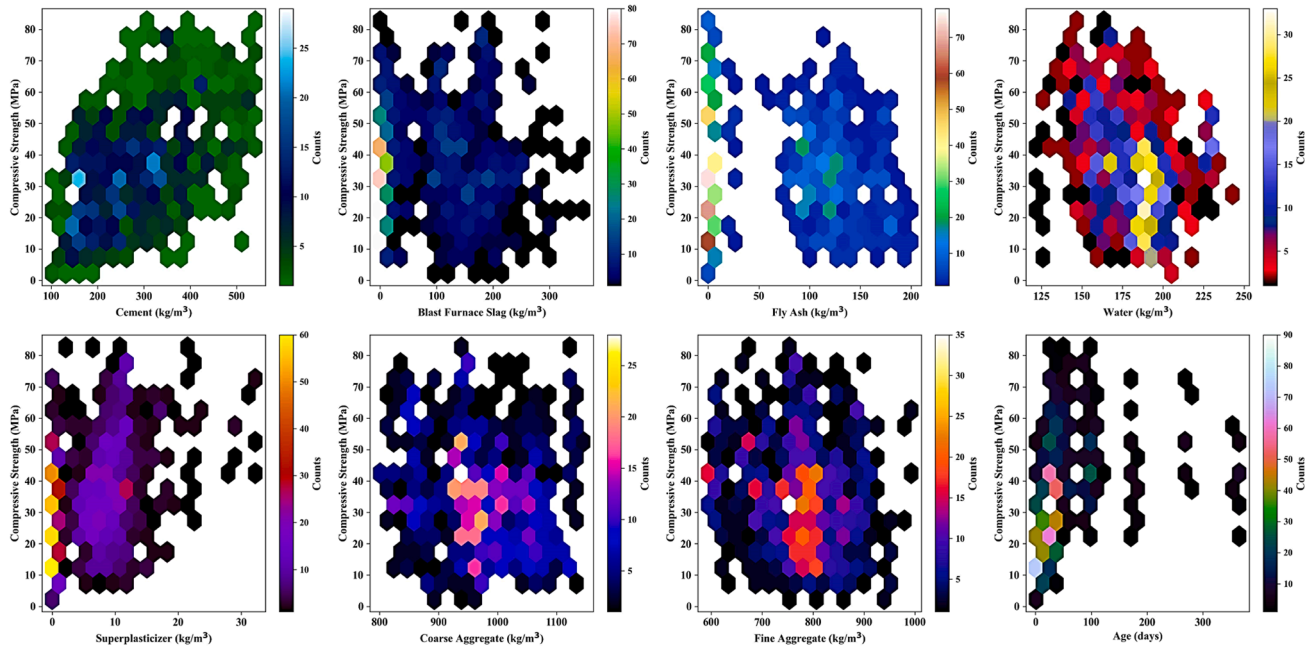


Fig. 4. Hexbin graphs of the relationship between input variables and CS of concrete.

The coarse aggregate (Cagg) plot shows a moderate positive correlation with CS. The data points are concentrated around the middle to higher ranges of Cagg. This suggests that a larger proportion of Cagg tends to improve CS, likely due to better packing and structural integrity of the concrete mix. The upper-middle-range of Cagg shows the densest data points, emphasizing its role in enhancing strength. The hexbin plot for fine aggregate (Fagg) shows a weak positive correlation with CS. The data points are scattered widely, with low density at higher ranges of Fagg. The concentration of points in the lower-middle ranges suggests that Fagg has a minor effect on CS, especially in comparison to other components such as cement or coarse aggregate. Lastly, the curing duration (Age) plot shows a clear positive correlation with CS, as expected. The data points form a tight upward trend, with a high concentration in the higher ranges of (Age), particularly in the upper-middle to higher ranges. This indicates that longer curing durations are directly associated with increased CS, confirming that prolonged curing allows concrete to reach its full potential strength. These insights suggest that optimizing cement and curing duration, along with moderate levels of slag and superplasticizer, can significantly enhance CS, while excessive fly ash and fine aggregate may offer diminishing returns.

4.5. Data normalization

When input data is of varying sizes, some ML models might not perform as well as they could. The significantly varying magnitudes of the various input features are shown in Table 2, where the cement ranges from 102 to 540 kg/m³, and the superplasticizers range from 0 % to 32.2 %. To fix this, we scale all the input variables to the same value using data normalization or rescaling. The max–min mapping function is used in this process, as shown in Eq. (2).

$$X_n = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (2)$$

Every input variable in this equation has a minimum (X_{min}) and maximum (X_{max}) value, and X_n represents the normalized data. The original dataset that is being rescaled is denoted by X . The main advantage of data rescaling is that it can speed up computations and make the ML prediction model more accurate and stable.

4.6. Description of base models

4.6.1. ANN model

The concept of Artificial Neural Networks (ANNs), inspired by the neural networks in the human brain, was first developed by McCulloch and Pitts in 1943 [81]. ANNs are adept at modeling linear and nonlinear systems without the constraints often associated with traditional statistical methods. Typically, an ANN comprises three layers: an input layer, one or more hidden layers, and an output layer. The training process of an ANN involves adjusting connection weights according to a learning rule derived from an input–output dataset, with the goal of minimizing the difference between predicted outputs and actual values [82]. This training process includes two main phases: feed-forward and back-propagation. During the feed-forward phase, data flows from the input layer through the hidden layers to the output layer. In the back-propagation phase, weights are adjusted from the output layer back to the input layer based on the overall error in the network. To further reduce this error, an appropriate back-propagation algorithm is employed, with the Levenberg-Marquardt technique being the most efficient and commonly used method for this purpose. Fig. 5 shows that the eight variables were used as inputs to the ANN model, and the output variable Y , which represents the concrete CS, was set as the outcome.

4.6.2. RF model

The Random Forest (RF) model, initially introduced by Ho [83] and further developed by Breiman [84], is an advanced ensemble learning method that utilizes multiple decision trees to enhance prediction accuracy. This technique applies bootstrap resampling to create various instances from the original dataset, with each instance used to construct an individual decision tree [85]. Predictions from these trees are then averaged to produce the final output. RF improves on the limitations of traditional decision trees, such as susceptibility to overfitting, by incorporating randomness in the selection of parameters during the tree construction. This not only makes the model more robust by combining well-established trees but also balances the weaknesses and strengths across different classes. The model structures condition hierarchically from the root to the terminal nodes, with decision-making processes at each node and a simple regression model at each terminal node to make predictions. Pruning is applied to reduce complexity and enhance the model's generalization capabilities. Fig. 6 shows the RF model's

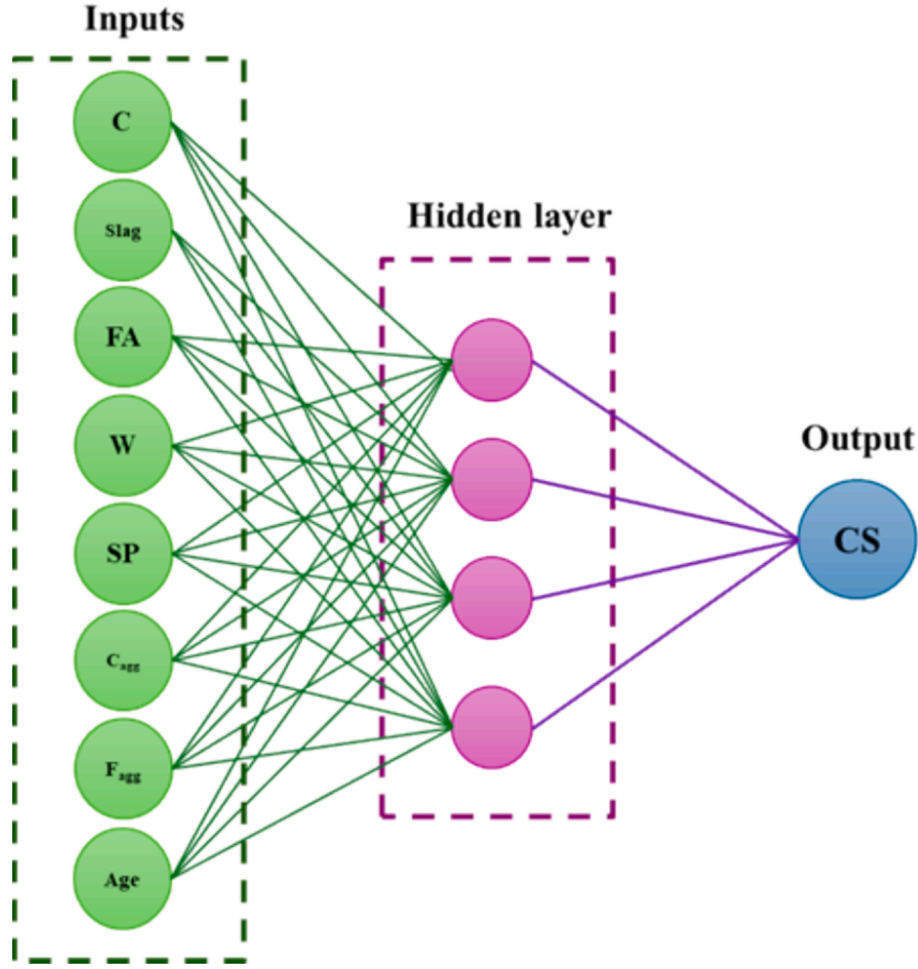


Fig. 5. Schematic representation of the ANN model development.

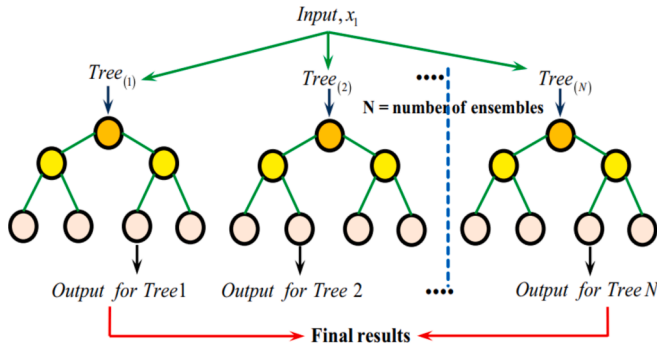


Fig. 6. Architecture of RF model [86].

architecture.

4.6.3. XGB model

The XGB model, renowned for its implementation of gradient-boosted decision trees, is specifically engineered to achieve exceptional speed and performance. The algorithm is characterized by its exceptional flexibility and versatility, making it particularly adept at managing sparse data and efficiently performing both regression and classification tasks. Its scalability and capacity to generate very competitive predictive models have enabled its widespread use in many ML competitions [85]. This study employs XGB due to its beneficial features, including regularization to mitigate overfitting and the use of

second-order gradients to accelerate convergence. Additionally, XGB is proficient at handling missing data while making split decisions and incorporates stochastic gradient descent to enhance model diversity and further mitigate overfitting. Moreover, XGB includes system-level improvements, including cache optimization and parallel processing, which contribute to its computational efficiency and its ability to manage large datasets effectively. The architecture of the XGB model is depicted in Fig. 7.

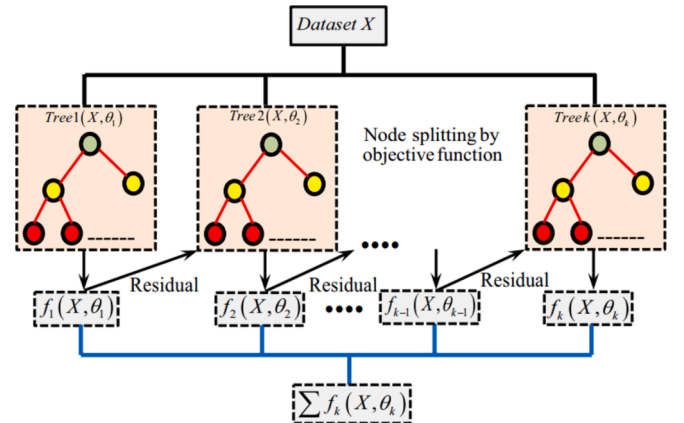


Fig. 7. Architecture of XGB model [87].

4.7. Hyperparameters tuning

In this study, Bayesian Optimization (BO) is employed to optimize hyperparameter settings, offering a more efficient alternative to traditional methods like Grid Search (GS) and Random Search (RS) [88]. Unlike GS, which exhaustively searches the entire hyperparameter space, BO models the objective function's prior distribution and refines the search to identify the optimal configuration. The BO process begins with defining the hyperparameter ranges and initial distributions for each model, aiming to maximize performance within these predefined bounds. By reducing unnecessary trials, BO accelerates the identification of the most effective hyperparameter combinations, significantly enhancing tuning efficiency [89]. Critical hyperparameters, such as the number of estimators, learning rate, maximum tree depth, and regularization parameters, were optimized using BO. This systematic approach minimizes trial-and-error, ensuring that the models achieve optimal performance. The selected configurations were validated on the test dataset to confirm their predictive efficacy.

Furthermore, this research integrates k-fold cross-validation (CV) to assess model robustness and mitigate overfitting. In particular, a 10-fold CV is utilized, where the dataset is partitioned into ten subsets, with each serving as the test set while the remaining nine are used for training. This method, known for balancing computational efficiency and the bias-variance trade-off, was combined with BO for hyperparameter optimization (BO + 10CV). The use of a 10-fold CV ensures accurate performance estimation while maintaining manageable computational demands.

4.8. Description of the stacked linear regression model

Stacking is an ensemble learning technique designed to enhance the predictive accuracy of machine learning models by combining multiple individual models [90,91]. The fundamental idea is to train several base models (BMs), often referred to as level-0 models, on the same dataset and then use another model, known as the *meta-model* or level-1 model, to aggregate their predictions into a final prediction. In a stacking ensemble, multiple BMs are trained independently on the training data. These models make predictions on the validation dataset, generating several forecasts. These predictions form a new dataset, where each column represents the predictions from a different base model. The *meta-model* is then trained on this new dataset to learn how to combine the base model predictions into a final output.

When linear regression is used as the *meta-model* in a stacking ensemble, the process starts with training the base models: ANN, RF, and XGB on the training dataset. These BMs make predictions on the validation dataset, creating a new dataset composed of these predictions. A linear regression model is trained on this new dataset, using the actual target values from the validation data as the response variable. The linear regression model learns to weigh the contributions of each base model's predictions to produce a final, combined prediction.

Stacking with a linear regression *meta-model* can significantly improve model performance by leveraging the strengths of various BMs. This method allows for flexibility in model choice and can capture diverse patterns in the data, making the final model more robust and accurate. Additionally, by combining multiple models, stacking reduces the risk of overfitting, as the *meta-model* balances the strengths and weaknesses of each base model. By employing stacking with linear regression, we achieve a more accurate and robust predictive model that benefits from the diverse insights provided by the base models. This approach is particularly effective in complex datasets where no single model performs optimally.

4.9. Evaluation methods of adopted models

The dataset was carefully divided into three separate sets: training, validation, and testing, with respective weights of 80 %, 10 %, and 10 %.

Both the training and validation datasets were used to develop and test the BMs. The SM, on the other hand, was built using validation-stage BM predictions. Then, the remaining 10 % of the database was used to test its performance. This way, we can test the model thoroughly on data it has never seen before while still training it on a statistically valid subset of the data. However, when comparing and evaluating the adopted models, two popular approaches are quantitative and visual methods [92].

Visual methods, such as scatter plots, violin boxplots, and Taylor diagrams, provide quick insights into the accuracy of various statistical measures, including maximum, minimum, median, and quartiles [93,94]. These visual tools offer rapid, engaging, and informative, easily interpretable comparisons. However, they may not provide detailed information about overall model performance [95,96]. Seven quantitative metrics were employed to provide detailed numerical insights into model performance. These performance metrics equations and their ideal values are listed in Table 3.

The A20 index serves as a crucial metric for assessing the reliability and accuracy of predictive models [97,98]. The metric quantifies the percentage of predictions that lie within a 20 % error margin of the actual values, providing a straightforward assessment of the model's performance. The A20 index is calculated as follows.

$$A20 = \frac{m_{20}}{n} \quad (3)$$

where m_{20} represents the number of predictions where the predicted value is within 20 % of the actual value. A lower A20 value indicates better predictive reliability, as it suggests that a greater proportion of the model's predictions are close to the actual values.

4.10. Interpretation of studied features

To evaluate and interpret machine learning models at both broad and detailed levels, researchers employ the SHapley Additive exPlanations (SHAP) method, which is grounded in cooperative game theory principles [99]. This method is used to assess the comparative influence of input variables on the prediction process, providing a sophisticated tool within the field of explainable AI. SHAP facilitates the clarification of the intricate relationships between input variables and estimates made by models, offering critical insights into which features most significantly influence the predictions and how they do so [100].

On the other hand, the current study examined a Partial Dependence Plot (PDP) to illustrate the functional link between a limited set of input parameters and the model's predictions. It illustrates the extent to which

Table 3
Equations of performance metrics and their ideal values.

Metric	Equation	Ideal Value
Determination coefficient (R^2)	$R^2 = \frac{\sum_{i=1}^n (y_i - y_{mean})^2 - \sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - y_{mean})^2}$	1
Root Mean Square Error (RMSE)	$RMSE = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}}$	0
Mean Absolute Error (MAE)	$MAE = \frac{\sum_{i=1}^n y_i - \hat{y}_i }{n}$	0
Mean Absolute Percentage Error (MAPE)	$MAPE = \frac{\sum_{i=1}^n \frac{ y_i - \hat{y}_i }{y_i}}{n} \times 100$	0
Mean Bias Error (MBE)	$MBE = \frac{\sum_{i=1}^n (\hat{y}_i - y_i)}{n}$	0
Cumulative Prediction Error Index (CPI)	$CPI = \frac{\sum_{i=1}^n \hat{y}_i - y_i }{\sum y_i}$	0
Variance Accounted For (VAF)	$VAF = 1 - \frac{Var(y_i - \hat{y}_i)}{Var(y_i)}$	1
Where n is the dataset number y_i and $\hat{y}_i$ are actual and predicted i^{th} values, respectively.		

the predictions are influenced by the values of these specific input parameters [101]. Moreover, PDPs highlight the impact of each parameter on the predicted outcomes generated by a machine learning algorithm. PDPs consider all instances within the dataset as a global method, revealing the overall relationship between a feature and the forecasted outcome [102]. These plots offer valuable insights into the relative contribution of each input variable to the predicted outcome. Furthermore, one-dimensional PDPs (PDPs-1D) depict the relationship between the expected result and a single input parameter.

4.11. Graphical user interface (GUI)

A user-friendly GUI has been developed to make the predictive model for estimating CS easily accessible. This interface, built using Python's Tkinter library [103], leverages its simplicity and standard distribution for creating interactive applications [104]. The development process involves setting up the Python environment, with the design focused on user guidance and ease of use. Key components include input fields (TextEntry widgets) for user data entry, a Button widget to trigger predictions, and a Label or Text widget to display the predicted CS. The predictive model is seamlessly integrated into the GUI using the pickle library to load and apply the model based on user inputs. To facilitate accessibility, collaboration, and distribution, the GUI is hosted on GitHub, providing version control and easy sharing.

5. Results and discussions

5.1. Optimal hyperparameters of developed BMs

Hyperparameters play a crucial role in the performance and efficiency of machine learning models. Unlike model parameters, which are learned from the data during training, hyperparameters are set prior to the training process and guide the learning process. The selection of appropriate hyperparameters can significantly influence the model's ability to generalize and make accurate predictions on unseen data. For the ANN, the optimal configuration includes three hidden layers, each with 20 neurons. Both the hidden layers and the output layer use the sigmoid activation function, which helps achieve non-linearity and ensures the model can capture complex patterns in the data.

The RF model, optimized for this study, consists of 313 estimators, with each tree having a maximum depth of 23. The model uses a minimum samples split of 2.0 and a minimum samples leaf of 1.0, ensuring that each split and leaf has a sufficient number of samples to avoid overfitting. The XGB model is fine-tuned with 493 estimators and a learning rate of 0.067, balancing the speed of learning and convergence. The model's maximum depth is set to 4.0, allowing it to capture interactions between features. Additionally, a column sample with tree value of 0.724 and a sub-sample value of 0.40 are used to prevent overfitting by adding randomness to the model training process. These optimized hyperparameters enhance the performance and accuracy of the respective ML models, enabling them to deliver robust and reliable predictions.

5.2. Quantitative assessment of BMs

5.2.1. ANN model

Fig. 8 presents a scatter plot comparing actual versus predicted values using an ANN model. The plot includes data points for both the training (TR) stage, represented by blue hexagons, and the validation (VD) stage, represented by orange hexagons. These points are distributed around the equality line (0 % error), which denotes a scenario where predicted values perfectly match actual values.

In the TR stage, the performance metric of the R²-value is 0.972, indicating a strong linear relationship between the actual and predicted values. The error metrics are as follows: RMSE of 2.776 MPa, MAE of 1.169 MPa, MAPE of 4.23 %, and MBE of 0.194 MPa, suggesting that the model exhibits good prediction accuracy with a slight overestimation bias. Additionally, the CPI is 0.033, and the VAF is 0.973, further highlighting the model's robust performance. In the VD stage, the R²-value decreases from 0.972 to 0.913, which still indicates a strong linear relationship. However, the RMSE increases to 5.270 MPa and the MAE to 3.625 MPa, suggesting that the model's predictions are less accurate in the validation set compared to the training set. The MAPE is 13.60 %, and the MBE is 0.107 MPa, indicating a reduced prediction bias. The CPI is 0.103, and the VAF is 0.913, reflecting good model performance on unseen data.

Additionally, Fig. 8 includes two dashed lines representing ± 10 % deviation from the equality line, which helps visually assess the model's

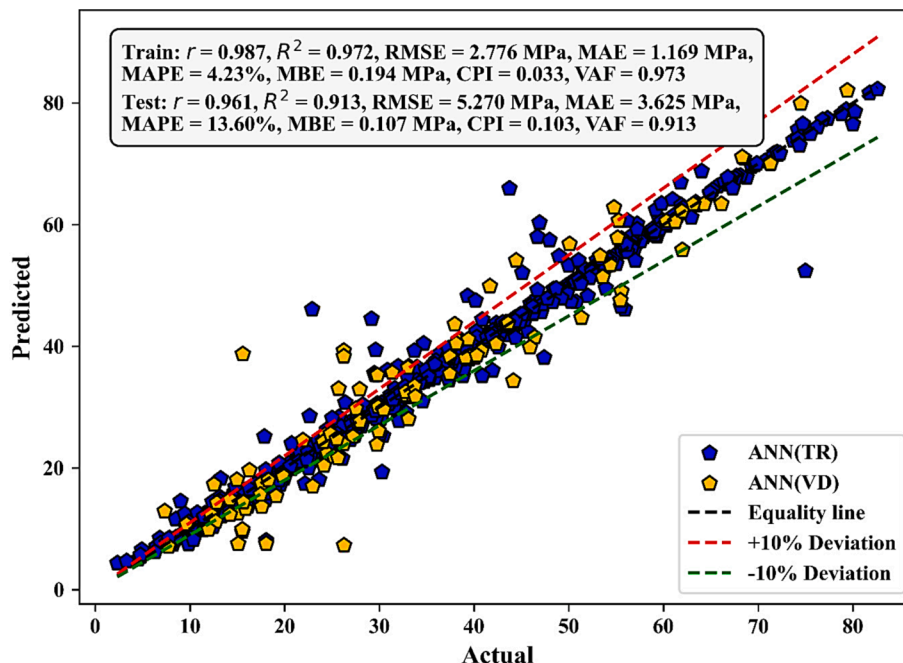


Fig. 8. Scatter plot comparing the actual and predicted CS-values based on the ANN model.

prediction accuracy. The majority of data points fall within these bounds, indicating that most predictions are within 10 % of the actual values. This visualization underscores the model's effectiveness in predicting values that closely match the actual data with a relatively small margin of error. Overall, the ANN model demonstrates strong performance on the training set and maintains good prediction accuracy on the validation set. The performance metrics indicate a robust linear relationship and low error rates, with most predictions falling within a ± 10 % deviation from the actual values.

5.2.2. RF model

Fig. 9 displays a scatter plot comparing actual versus predicted values using the RF model. This plot shares the same characteristics as the previous plot for the ANN model. During the TR stage, the R^2 -value is 0.985, indicating an extremely strong correlation between the actual and predicted values. The error metrics are RMSE = 2.049 MPa and MAE = 1.324 MPa, demonstrating the model's high prediction accuracy. Additionally, the MAPE and MBE values are 4.60 % and -0.077 MPa, respectively, indicating a slight underestimation bias. The CPI is 0.037, and the VAF is 0.985, further underscoring the model's strong performance.

In the VD stage, the R^2 value is 0.932, slightly lower than in the TR stage. The RMSE and MAE values are 4.350 MPa and 3.118 MPa, respectively, suggesting that the RF model's prediction errors have increased compared to the TR stage. The CPI is 0.088, and the VAF is 0.932, reflecting solid performance on unseen data. Moreover, most predictions fall within ± 10 % deviations from the equality line, highlighting the model's effectiveness in accurately predicting values with a relatively small margin of error.

Several observations can be made when comparing the RF model with the previous ANN model. Both models exhibit strong linear relationships between actual and predicted values, with high correlation coefficients and R^2 values in both the TR and VD stages. However, the RF model shows slightly higher correlations during the TR and VD stages, with an increase in R^2 of 1.30 % and 1.90 %, respectively, compared to the ANN model. This indicates that the RF model captures the variance in the TR stage slightly better than the ANN model. Regarding the error metrics, the RF model has lower RMSE and MAE values than the ANN model in the TR stage, with the ANN model being higher by 0.727 MPa

and 0.155 MPa in RMSE and MAE, respectively. This suggests that the RF model had better prediction accuracy. In the VD stage, the RF model also shows a lower RMSE by 0.92 MPa compared to the ANN model and a slightly lower MAE by 0.245 MPa. This indicates that the RF model has a lower overall error, while the ANN model may be more susceptible to larger individual errors. Finally, both the RF and ANN models have similar MAPE values during the TR stage (RF = 4.60 % and ANN = 4.23 %). However, when tested with unseen data in the VD stage, the RF model shows a lower MAPE, with a difference of 1.86 % compared to the ANN model.

In summary, while both RF and ANN models perform well, the RF model performs slightly better in capturing variance and minimizing errors, particularly in the VD stage. The ANN model, though robust, shows marginally higher errors and lower correlation metrics, especially in the VD stage.

5.2.3. XGB model

Fig. 10 presents a scatter plot comparing actual versus predicted values using the XGB model. This plot shares the same features as the previous ones for the ANN and RF models. During the TR stage, the R^2 -value indicates an exceptionally strong linear relationship between the actual and predicted values, demonstrating that the model's predictions closely align with the actual values. The RMSE (2.002 MPa), MAE (1.384 MPa), and MAPE (5.05 %) are relatively low, suggesting that the XGB model exhibits a low average error. However, the errors increase slightly in the VD stage compared to the TR stage, indicating possible overfitting. The CPI is 0.038, and the VAF is 0.986, further highlighting the model's strong performance.

In the VD stage, the R^2 -value shows a minimal decrease from 0.986 to 0.953, indicating a good fit. The RMSE and MAE values of 3.859 MPa and 2.715 MPa, respectively, suggest that the XGB model's prediction errors have slightly increased compared to the TR stage. The MAPE was 9.58 %, and the MBE was -0.070 MPa, indicating a slight prediction bias. The CPI is 0.077, and the VAF is 0.953, reflecting good model performance on unseen data. Additionally, most data points fall within the ± 10 % bounds, indicating that the majority of predictions are within 10 % of the actual values. This visualization underscores the model's effectiveness in predicting values that closely match the actual data with a relatively small margin of error. Several observations can be

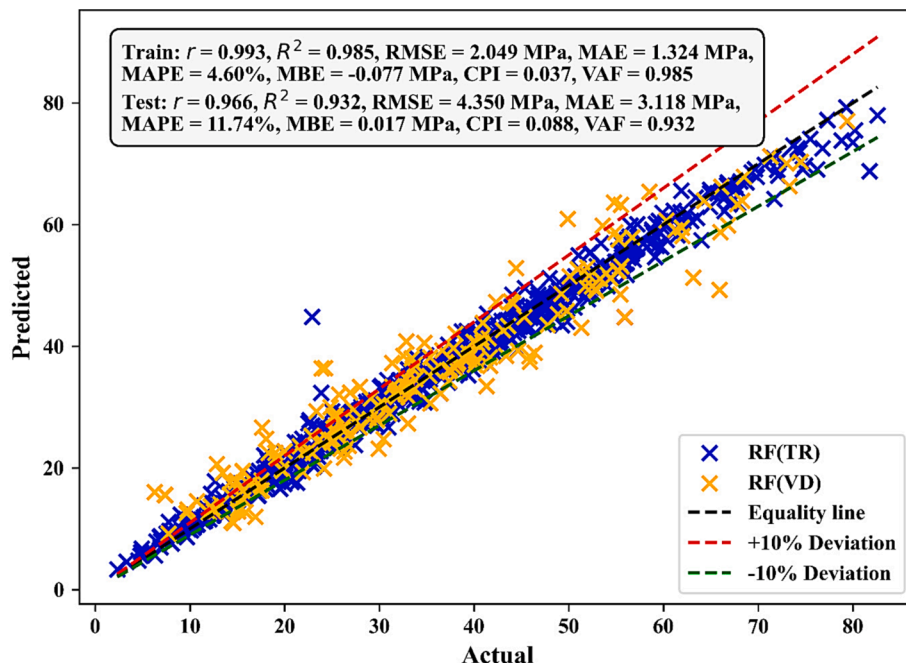


Fig. 9. Scatter plot comparing the actual and predicted CS-values based on the RF model.

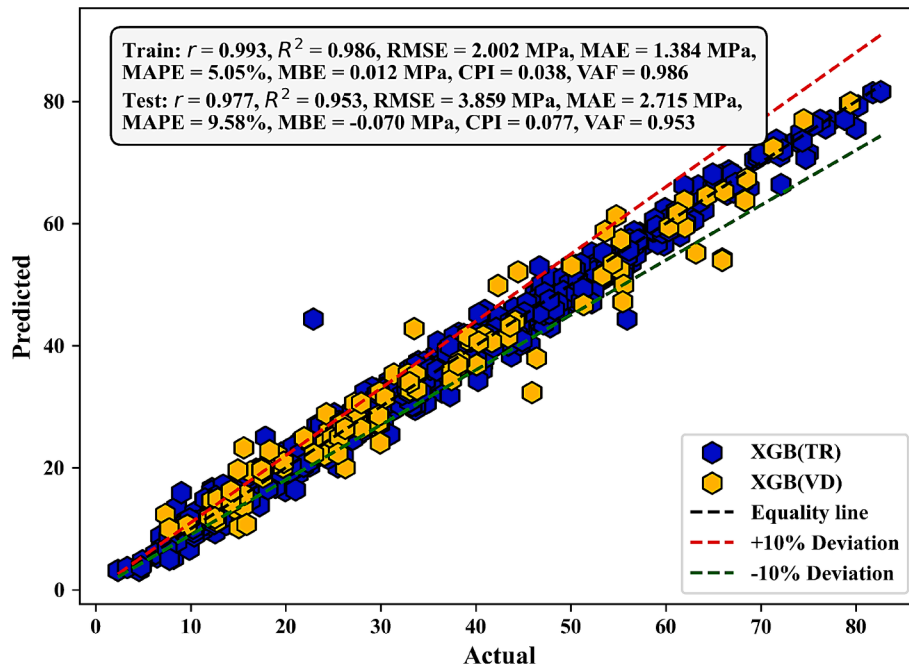


Fig. 10. Scatter plot comparing the actual and predicted CS-values based on the XGB model.

made when comparing the XGB model with the previously discussed ANN and RF models:

1. All three models demonstrate strong linear relationships between actual and predicted values, with R^2 -values greater than 0.91 during both the TR and VD stages.
2. The XGB model, followed by RF, has slightly higher R^2 values than the ANN model, suggesting a better capture of variance in both the TR and VD stages.
3. The RF and XGB models exhibit similar RMSE and MAE values during the TR stage, which are lower than those of the ANN model, indicating better prediction accuracy. In the VD stage, the XGB model shows lower RMSE than both the ANN and RF models, indicating superior performance in these error metrics.
4. The MAPE values for the RF and XGB models are approximately the same during the TR stage and are lower than those for the ANN model. In the VD stage, the XGB model exhibits lower MAPE than both the ANN and RF models, suggesting that the XGB model may generalize better in terms of MAPE.

5. The MBE for all models indicates minimal prediction bias, with the XGB model displaying the least bias. The CPI and VAF metrics consistently indicate strong model performance across all models, with the RF and XGB models slightly outperforming the ANN model.

In summary, the XGB model demonstrates slightly better overall performance than the others. The RF model performs comparably, with substantial prediction accuracy and minimal bias. However, The ANN model shows a larger deviation, suggesting it may be more prone to overfitting than the other two models.

5.3. Visual assessment of BMs

5.3.1. Violin plots

The violin plots provide a visual comparison of the distribution of predicted values from various models against the actual values. A model whose predicted values closely match the shape and spread of the actual values' violin plot is considered more accurate. These plots also help identify significant deviations or biases in the model predictions. Fig. 11

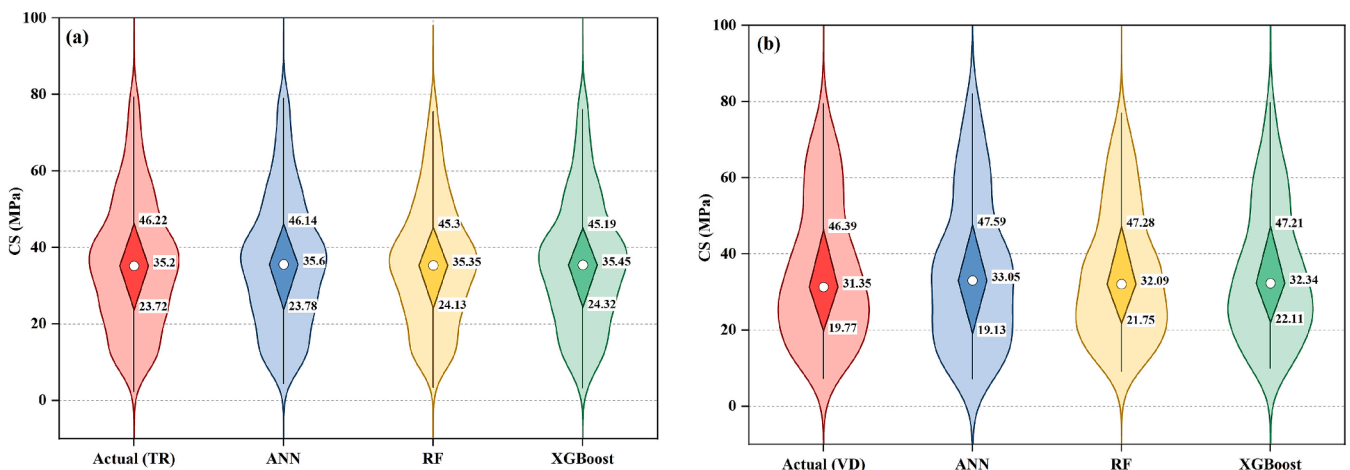


Fig. 11. Violin plots comparing the predicted CS-values from the BMs during the (a) TR and (b) VD stages.

features two violin plots comparing the predicted CS-values from different BMs during the TR and VD stages. The CS values are plotted along the y-axis, while the x-axis represents various models. Each violin plot displays the kernel density estimation of the data distribution, with wider sections indicating a higher density of data points. The central white dot signifies the median value, while the thick band encompassing the upper and lower values denotes the interquartile range (IQR).

In the TR stage (Fig. 11a), the RF and XGB models are identified as the top-performing models, exhibiting centered medians of 35.35 and 35.45, respectively, which closely align with the actual CS of 35.20 MPa. The plots exhibit symmetry relative to the actual data, suggesting minimal variance. The interquartile ranges are 24.13 to 45.30 MPa for the RF model and 24.32 to 45.19 MPa for the XGB model, both closely aligning with the actual CS range of 23.72 to 46.22 MPa. The compact distribution indicates that the RF and XGB models exhibit a strong fit to the training data. The ANN model demonstrates slightly broader yet still well-structured distributions. The data exhibited a median of 35.60 and an IQR from 23.78 to 46.14 MPa, showing moderate variance alongside reliable performance.

During the VD stage (Fig. 11b), the XGB and RF models exhibit superior performance, characterized by narrow and symmetric distributions that effectively generalize from the training data, with XGB showing a slight advantage over RF. The XGB model exhibits a median of 32.34 MPa with an IQR spanning from 22.11 to 47.21 MPa. The RF model shows a median of 32.09 MPa and an IQR from 21.75 to 47.28 MPa. Both models demonstrate low variance relative to the actual CS values, which have a median of 31.37 MPa and an IQR from 19.77 to 46.39 MPa. The ANN model demonstrates greater variance, presenting a median of 33.05 MPa and an interquartile range from 19.13 to 47.59 MPa, indicating reduced generalization capability.

5.3.2. Taylor diagrams

The second visual assessment involved using Taylor diagrams, which graphically compare the statistical properties of different models with the actual data. These diagrams feature axes for standard deviation (SD) and correlation coefficient (CC), with the actual data represented as a reference point. The radial distance from the origin indicates the SD, the angle represents the CC, and the distance from the actual point reflects the centered Root Mean Square Difference (RMSD) between the model predictions and the actual values. Fig. 12 displays the Taylor diagrams comparing predicted CS-values from the BMs during the TR and VD stages.

The statistical results, including SD and CC, are derived from these diagrams.

For the ANN model, the SDs are 16.94 MPa in the TR stage and 19.15 MPa in the VD stage, with corresponding CCs of 0.986 and 0.967. The RF model exhibits SDs of 16.10 MPa in the TR stage and 16.70 MPa in the VD stage, with CCs of 0.993 and 0.972, respectively. The XGB model shows SDs of 16.50 MPa in the TR stage and 17.35 MPa in the VD stage, with CCs of 0.993 and 0.976. The results clearly indicate that the RF and XGB models outperform the others in terms of SD and CC values. The XGB model, in particular, has the highest CC and the smallest centered RMSD, demonstrating its superior ability to capture the variance and trends in the data accurately. The RF model, while slightly trailing the XGB model, still performs strongly with a high CC and low centered RMSD. The ANN model, however, exhibits a higher centered RMSD and slightly lower CC than the XGB and RF models, suggesting that although it captures the general trends, it is less precise in matching the actual data distribution and has a larger prediction error. Thus, the visual assessments conducted highlight the efficacy of the XGB and RF models in accurately and consistently predicting CS-values.

5.3.3. Rank analysis

A rank analysis was conducted to evaluate the overall performance of the developed base models using the performance metrics presented in Table 3. In this analysis, a model attaining an overall rank value of 1 signifies the highest level of performance, whereas a rank of 3 denotes the lowest level of performance. The sum of the individual ratings across various metrics is used to determine the overall ranking of each model. The model exhibiting the greatest cumulative rank is regarded as the least efficient, while the model exhibiting the smallest cumulative rank is defined as the most efficient. An analysis of the rankings for both the TR and TS stages is presented in Table 4.

During the TR stage, the XGB model emerges as the most exceptional performer, achieving a total score of 13 by consistently ranking first across all seven performance metrics. The RF model follows closely with a total score of 14, ranking second overall and demonstrating strong performance. The ANN model secures the third spot with a score of 15. All models developed in the TR stage exhibited robust predictive capabilities, as evidenced by their close scores. In the TS stage, the XGB model emerges as the most effective, achieving the lowest total score of 8, indicating top performance across all metrics. This consistency from the training to the testing stages underscores its robustness and

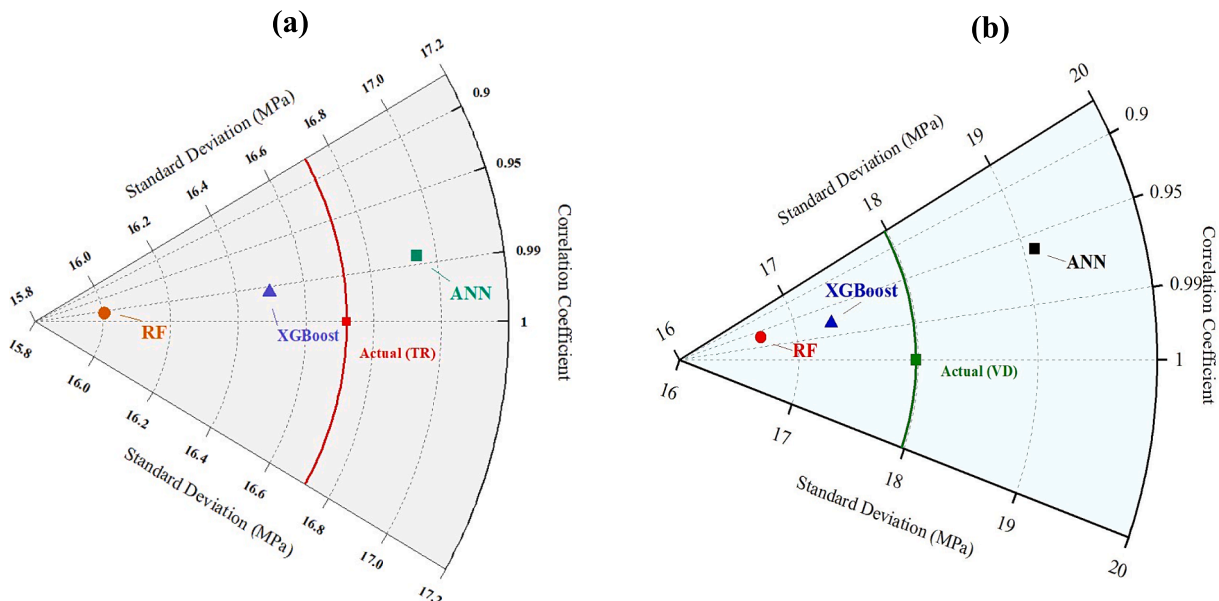


Fig. 12. Taylor diagrams comparing predicted CS-values from the BMs during the (a) TR and (b) VD stages.

Table 4

Rank analysis for the developed BMs.

Model	Stage	R ²	RMSE (MPa)	MAE (MPa)	MAPE	MBE	CPI	VAF	Total Score	Overall Rank
ANN	TR	3	3	1	1	3	1	3	15	36
	VD	3	3	3	3	3	3	3	21	
RF	TR	2	2	2	2	2	2	2	14	27
	VD	2	2	2	2	1	2	2	13	
XGB	TR	1	1	3	3	1	3	1	13	21
	VD	1	1	1	1	2	1	1	8	

reliability. The RF model follows in second place with a score of 13, demonstrating dependable performance in real-world testing scenarios. The ANN model remains the least effective in the testing phase, with a score of 21, consistent with its training performance, highlighting significant challenges in predictive accuracy. Overall, the XGB model distinguishes itself among three BMs, achieving an impressive overall score of 21. The RF model holds the second rank with a score of 27. Lastly, the ANN model consistently ranks lowest, with notably high scores of 36, indicating the need for further improvement in its predictive capabilities.

5.4. Prediction of concrete CS via SM-LR

The prediction of concrete CS using a Stacked Model-based Linear Regression (SM-LR) involves integrating the outputs of three distinct base models: ANN, RF, and XGB. Through a systematic trial-and-error process, a linear equation was derived that optimally combines the predictions from these individual models into a cohesive ensemble model. The equation for the SM-LR model is as follows:

$$CS_{(SM-LR)} = -0.7122 + 0.1877 CS_{(ANN)} + 0.4506 CS_{(RF)} + 0.3814 CS_{(XGB)} \quad (4)$$

In this equation, the coefficients associated with each BM indicate

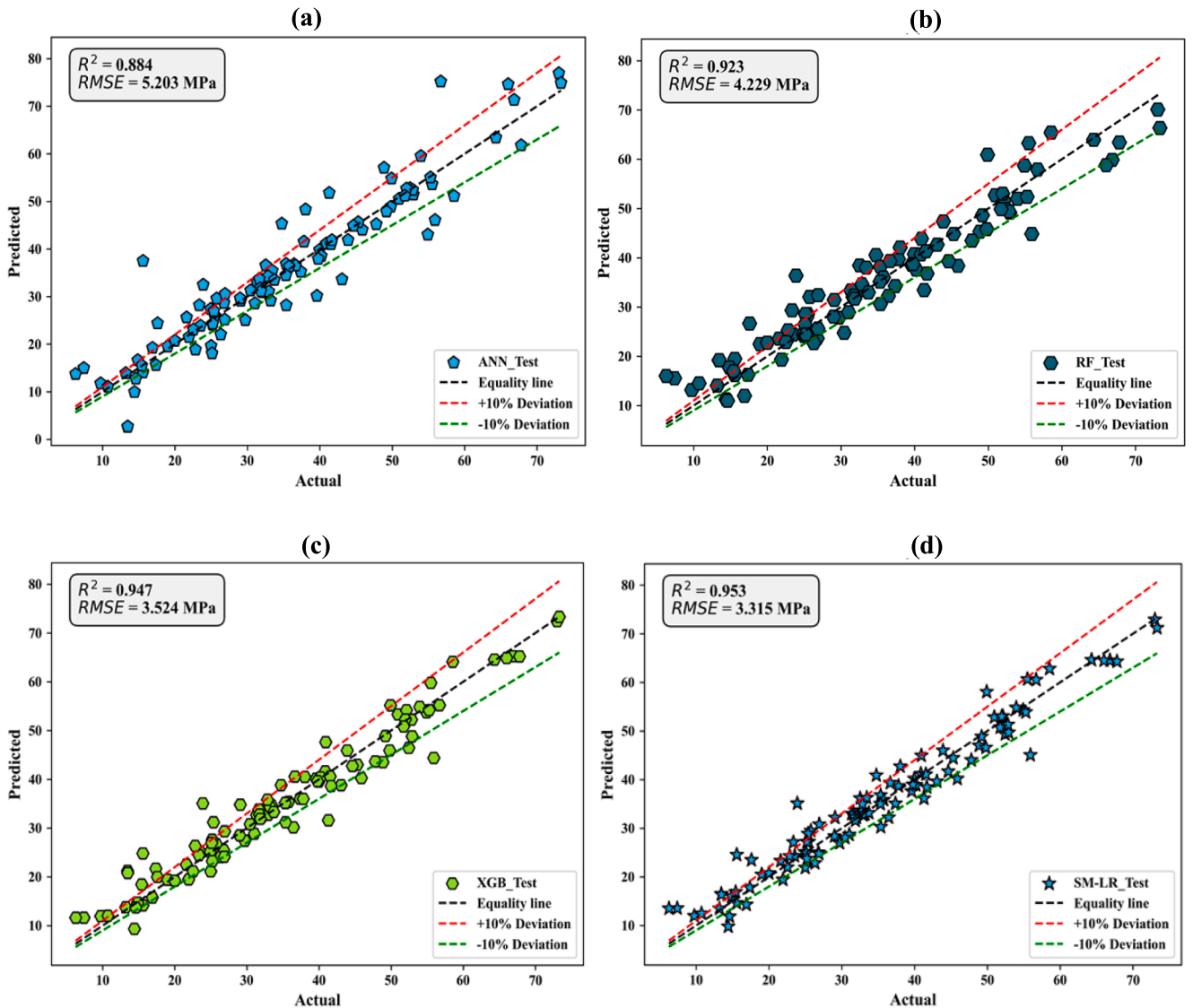


Fig. 13. Scatter plots during the TS stage for (a) ANN, (b) RF, (c) XGB, and (d) SM-LR models.

their relative contributions to the final prediction. The ANN model contributes 0.1877 to the overall prediction, RF provides the most substantial weight of 0.4506, and XGB adds a contribution of 0.3814. The constant term (-0.7122) adjusts the predictions for any inherent biases in the model.

5.5. Comparison between the BMs and SM-LR

5.5.1. Quantitative assessment

Fig. 13 presents scatter plots illustrating the performance of four models (ANN, RF, XGB, and SM-LR) in predicting CS values during the testing (TS) stage. Each plot compares actual CS values with predicted values, with individual data points representing specific predictions. The plots include a dashed black line representing perfect predictions (equality line) and dashed red and green lines indicating $\pm 10\%$ deviation from the equality line. The models' performance was evaluated using the R2 and RMSE metrics displayed on each plot.

The ANN model (Fig. 13a) demonstrates the lowest performance among the four models, with an R2-value of 0.884 and an RMSE of 5.203 MPa. While many data points fall within the deviation lines, significant deviations suggest that the ANN model's prediction accuracy could be improved. In contrast, the RF model (Fig. 13b) shows a higher R2-value of 0.923, indicating better predictive performance than the ANN model, and an RMSE of 4.229 MPa, reflecting a lower error rate and improved prediction accuracy. A more significant proportion of the RF model's data points lie within the deviation lines, underscoring its solid performance in accurately predicting CS values. The XGB model (Fig. 13c) achieves an R2-value of 0.947, indicating a very high correlation between actual and predicted values. The model's RMSE is 3.524 MPa, signifying a lower error rate and superior prediction accuracy compared to both the RF and ANN models. Data points for the XGB model are closely clustered around the equality line, highlighting its robust performance. The SM-LR model (Fig. 13d) outperforms the other models, with the highest R²-value of 0.953 and the lowest RMSE of 3.315 MPa, suggesting the best fit between actual and predicted values, as well as the smallest prediction errors. Most data points for the SM-LR model are tightly clustered around the equality line, underscoring its exceptional predictive accuracy.

Additionally, the A20 results for the tested models indicate significant differences in their predictive performance, allowing for an assessment of the reliability of the adopted models. The XGB model recorded the lowest A20 index at 10.07, indicating it provides the most accurate and reliable predictions compared to the other models. The SM-LR model exhibits a low A20 value of 10.08, indicating its reliability is nearly comparable to that of XGB. The two models are the most accurate and reliable for the TS dataset. The RF model exhibited moderate performance, achieving an A20 index of 13.07. Although this surpasses XGB and SM-LR, it still reflects a satisfactory level of prediction accuracy, albeit with somewhat increased variability in the predictions. The ANN model exhibited the highest A20 index of 13.48, indicating it was the least reliable among the four models assessed. This indicates that ANN exhibited the highest proportion of predictions exceeding the 20% error margin, demonstrating reduced consistency and precision relative to the other models. The A20 index results indicate that XGB and SM-LR yield the most accurate predictions with minimal error, whereas ANN and RF exhibit greater error rates, with ANN being the least dependable.

According to the aforementioned quantitative evaluation, the SM-LR model demonstrates the best performance during the TS dataset, followed closely by the XGB model, which also exhibits high predictive accuracy. The RF model performs well but with slightly lower accuracy than the XGB model. The ANN model, while still effective, shows the lowest accuracy among the four models. These results highlight the effectiveness of the stacking method used in the SM-LR model, which enhances prediction accuracy and solid model performance when applied to unseen data.

5.5.2. Visual assessment

Fig. 14a presents violin plots comparing the predicted CS-values from the base models and the SM-LR during the TS stage. The distribution of the SM-LR model closely matches the actual values, with the SM-LR exhibiting a median of 33.72 and an IQR from 24.58 to 44.95. In contrast, the actual testing dataset shows a median of 33.21 and an IQR from 24.92 to 45.85, suggesting robust predictive capability relative to all BMs employed. The XGB and RF models follow the SM-LR, exhibiting distributions that closely resemble the actual testing values. The XGB model has a median of 34.4 and an IQR spanning from 24.81 to 43.68, whereas the RF model has a median of 33.47 and an IQR range from 24.65 to 44.84. Unexpectedly, the ANN model demonstrates a reduced deviation in the median and IQR when compared during the TR and VD stages, with a median of 33.73 and an IQR spanning from 24.41 to 45.65; however, it remains inferior to the XGB and RF models in ranking.

Fig. 14b displays Taylor diagrams comparing the predicted CS-values among the base models and SM-LR in the TS stage. The ANN model shows an SD of 16.10 MPa and a CC of 0.946, while the RF model has an SD of 14.20 MPa and a CC of 0.955. The XGB model exhibits an SD of 14.60 MPa and a CC of 0.972, and the SM-LR model demonstrates an SD of 14.80 MPa and a CC of 0.984. These results indicate that the SM-LR model performs exceptionally well, with a high CC and low centered RMSD, closely followed by the XGB model. The RF model also shows strong performance but has a slightly lower correlation coefficient and a higher centered RMSD than the SM-LR and XGB models. The ANN model, however, shows the lowest CC and the highest centered RMSD among the four models, suggesting it is less precise in matching the actual data distribution and has a larger prediction error compared to the top-performing models.

In summary, quantitative analysis and visual plots confirm that the SM-LR model outperforms the other models by accurately capturing the statistical properties of the data and minimizing prediction errors. The XGB model closely follows the SM-LR model in performance, while the RF model demonstrates good accuracy but slightly higher errors. Although still effective, the ANN model shows comparatively lower accuracy and higher error rates.

5.5.3. K-fold CV analysis

A more accurate assessment of model efficacy is achieved through k-fold cross-validation (CV), which reduces the likelihood of overfitting to specific subsets of the data. By mitigating the risk associated with a single train-test split, k-fold CV provides a more robust evaluation of model performance [105]. The results indicate that the models exhibit both high accuracy and reliability. Fig. 15 illustrates a comparison of the developed models utilizing 10-fold across 10-fold CV using five performance metrics: R², RMSE, MAE, MAPE, and MBE. For the R² metric (Fig. 15a), all models show high R² values across all folds, with SM-LR and XGB consistently achieving values close to 1.0, indicating strong model performance. The RF model also performs well, while the ANN model shows slightly more variability in R²-values. For the RMSE metric (Fig. 15b), the SM-LR model exhibits the lowest RMSE values, highlighting its accuracy in minimizing significant errors. The XGB and RF models follow, with the RF showing slightly higher RMSE in some folds. The ANN model again shows slightly higher RMSE than the others, indicating it might be less robust to large errors.

The SM-LR model consistently has lower MAE and RMSE values across most folds (Fig. 15c). The XGB, ANN, and RF models follow closely, with the ANN sometimes outperforming others in specific folds. Lastly, MBE values closer to zero suggest unbiased predictions for MBE values. Accordingly, as shown in Fig. 15d, the XGB and SM-LR models have MBEs closer to zero across most folds, indicating minimal bias. The ANN model shows more variability, suggesting it might have some systematic bias depending on the fold. The RF model also displays some bias, though it varies significantly across folds. Overall, the SM-LR model demonstrates the best performance across most metrics, with

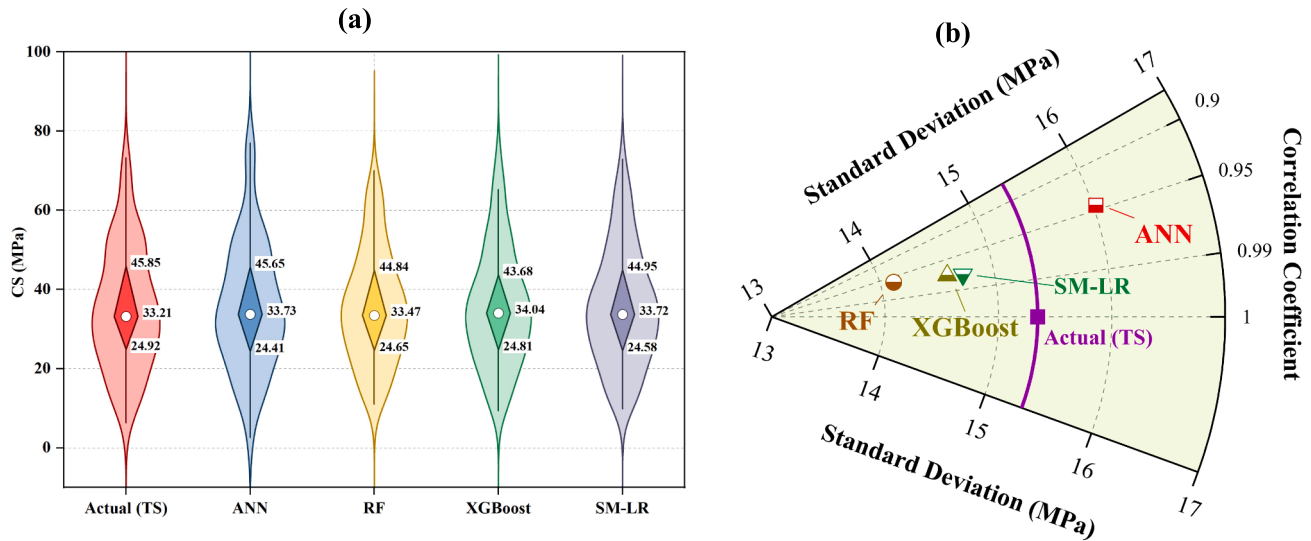


Fig. 14. Visual evaluations for predicted CS-values between the BM and SM in the TS stage by (a) Violin boxplot and (b) Taylor diagram.

consistently low RMSE, MAPE, MAE, and MBE values, as well as high R^2 values, indicating superior predictive power and minimal bias. The XGB and RF models also show good performance, particularly in R^2 and MAE, and can be considered reliable alternatives if computational efficiency or simplicity is a concern. Finally, the ANN model tends to show slightly worse performance in terms of error metrics but still provides a reasonably high R^2 , suggesting it can still capture a significant portion of the variance in the data.

5.6. Feature importance and interpretability

5.6.1. SHAP analysis

The purpose of this sensitivity analysis was to assess the impact of individual input features on the output variable in the prediction of concrete compressive strength. The SHAP feature importance method was utilized to analyze the impact of each input variable during the testing stage based on the most effective predictive base ML model (i.e., XGB model). Fig. 16 presents the results, involving both dot and summary bar plots that offer significant insights into feature importance. The interpretations of the SHAP analysis are as follows:

- Inputs are ordered by importance, with the most important inputs at the top.
- The position of the dots along the x-axis shows the impact of the input on the concrete's CS value. The positive values indicate a positive impact, while the negative values indicate a negative effect.
- Inputs with longer bars exert a greater influence on the output of the model.
- Color coding facilitates the comprehension of the input's value by assigning blue to low values and red to high values.

Fig. 16a illustrates a comprehensive SHAP dot plot, with each dot denoting a SHAP value corresponding to a particular instance within the dataset. The figure demonstrates that the curing age (Age) exerts the most significant influence on the concrete CS, with values ranging from -100 to 250 . The cement content (C) shows a comparable trend to curing age, serving as the second critical input, which highlights its advantageous impact on the enhancement of concrete strength. The SHAP values exhibit a range from -75 to 100 , where elevated values signify enhanced concrete strength. The inputs fine aggregate (Fagg) and blast furnace slag (Slag) exhibit a moderate influence, as indicated by SHAP values that are less dispersed compared to those of curing age and cement content. The SHAP values range from -60 to 70 for the Fagg

variable and from -45 to 50 for the Slag variable. In contrast, the Superplasticizer (SP) exhibits a less pronounced effect, as indicated by SHAP values that cluster near zero, with a range from -25 to 20 , implying a minimal influence on CS values. The coarse aggregate (Cagg) and fly ash (FA) exhibit neutral to weak effects on the model's predictions, as indicated by SHAP values clustering near zero, which implies a minimal influence on compressive strength within the analyzed range of values. Lastly, water (W) exhibits a less pronounced effect on the model's predictions, as indicated by SHAP values that are predominantly clustered around zero. This suggests that although they can impact concrete strength, their influence is less significant in comparison to other variables.

In addition, Fig. 16b indicates that curing age is the primary variable affecting the CS of concrete, with cement content and fine aggregates following as secondary and third factors, respectively. The curing age directly influences the hardening process of concrete, whereas the cement content is essential for bonding the mix. Consequently, these two factors are pivotal in predicting strength. Slag and fly ash, while significant, provide a more moderate influence, generally contributing to the enhancement of long-term strength and durability. Coarse aggregates contribute significantly, albeit to a lesser extent, while the impact of water and superplasticizers is minimal. The low ranking of water indicates that other factors, such as cement content, mediate its influence. The limited impact of superplasticizers highlights their function in enhancing workability rather than directly affecting strength. The primary factors influencing concrete strength in this model are curing age, cement content, and aggregate properties.

5.6.2. PDP analysis

Fig. 17 illustrates the PDPs for eight features that have been carried out with the XGB model. The plots illustrate the marginal effect of each feature on the model's predictions, demonstrating the changes in predicted output as the value of each feature changes, with all other features kept at unchanged values. The plot indicates a distinct positive correlation between the feature value of input cement (C) and the model's prediction. The predicted values of CS exhibit a steady increase with the rise in cement content. The data indicates that the cement variable serves as a significant positive predictor within the XGB model. The plot for the blast furnace slag (Slag) displays a relatively flat line, indicating that variations in the Slag have minimal to no impact on the predicted concrete CS values. In a similar manner, the plot for the input fly ash (FA) shows a flat line, further indicating that input FA, similar to Slag, has a minimal impact on the predicted CS values. The analysis of

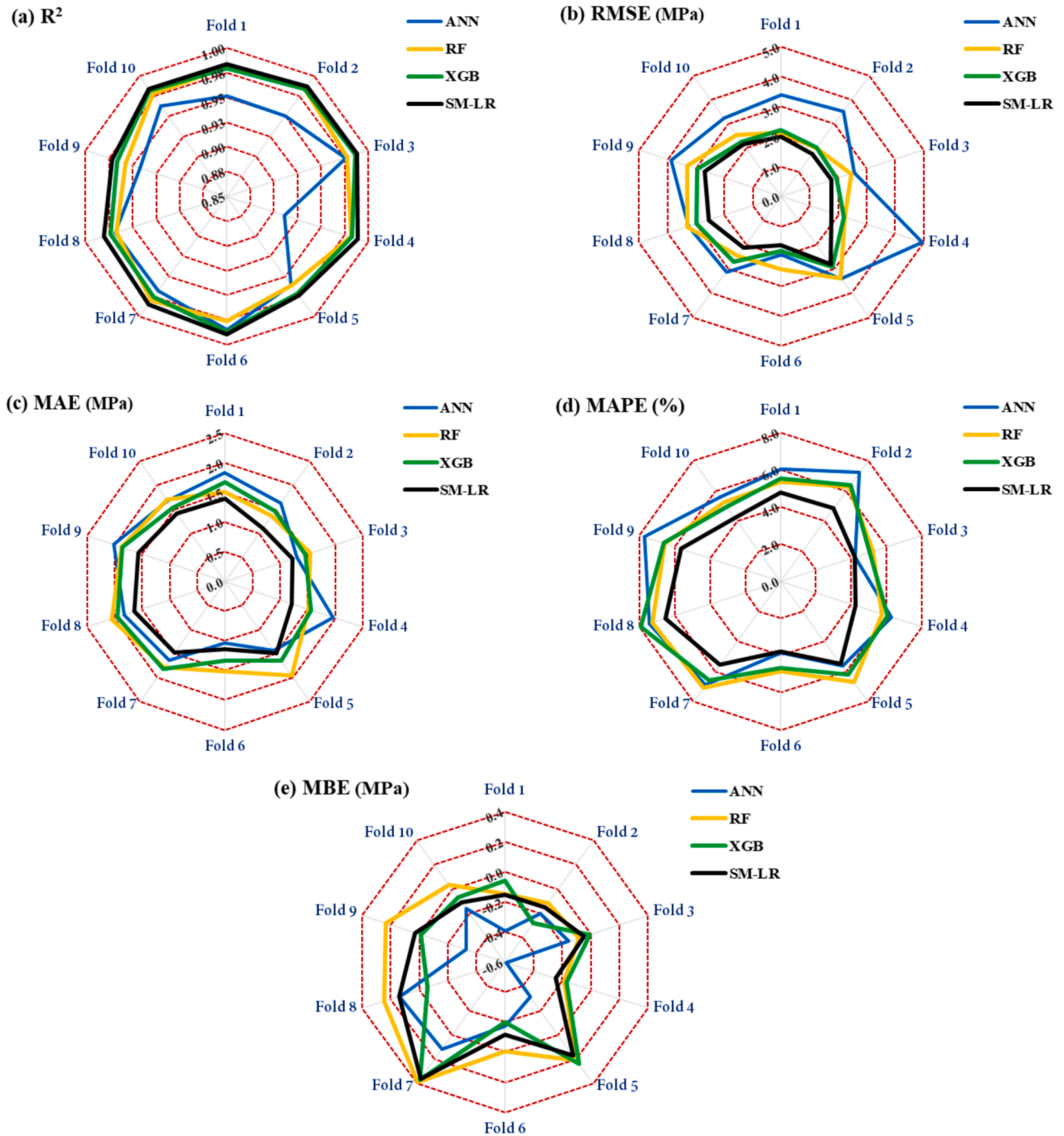


Fig. 15. Radar plots derived from BO + 10 folds CV across performance metrics: (a) R^2 , (b) RMSE, (c) MAE, (d) MAPE, and (e) MBE.

the water content (W) input indicates a negative correlation with the predicted values. As the water amount rises, the predicted values of CS decrease, which may indicate a risk factor or an adverse variable within the model. The plot of the superplasticizer (SP) shows minor fluctuations while generally trending upwards, suggesting a weak positive correlation with the predicted CS values. The influence of the SP on the predictions is positive, although not as significant as that of the cement content in the concrete mixture.

The plot for the coarse aggregate (Cagg) demonstrates a relatively flat trend with minimal fluctuations, suggesting that Cagg exerts negligible influence on the CS values. In a manner comparable to the Slag and FA variables, this indicates that the coarse aggregate does not play a significant role in the model's predictions and may be regarded as less

critical. The plot of fine aggregate (Fagg) displays fluctuations without a noticeable trend, indicating a weak or inconsistent correlation with the predicted CS values of concrete. The data indicates that the impact of fine aggregate on the model's predictions is neither strong nor consistent, making it a less reliable predictor. The plot of the curing duration (Age) presents a pronounced increase initially, followed by a stabilization phase. This demonstrates that, at the beginning, extending the curing duration resulted in notable enhancements in the concrete CS. However, beyond a specific threshold, additional increases yield only marginal improvements in predicted CS values.

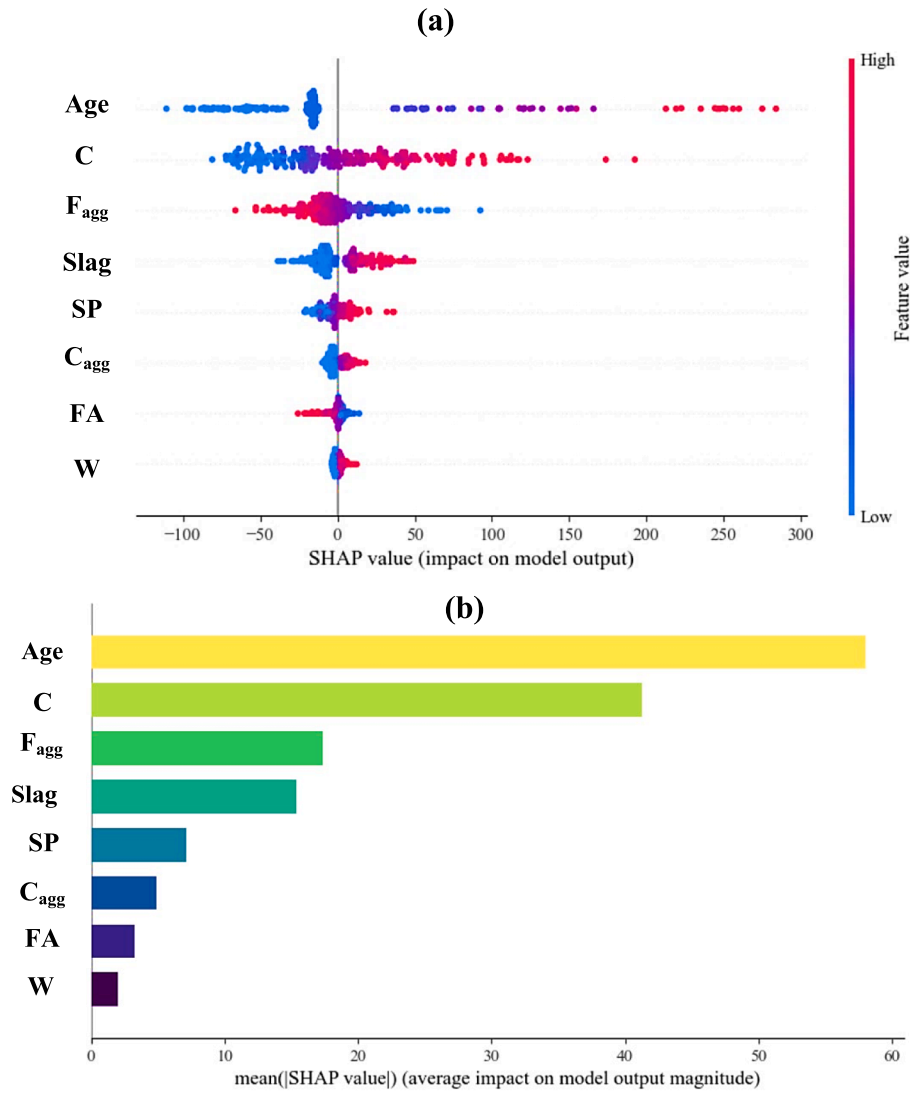


Fig. 16. SHAP charts: (a) summary dot plot and (b) feature importance bar plot based on the XGB model.

5.7. Comparison with previous studies

The developed SM-LR was evaluated against recent studies using ML and deep learning (DL) approaches to predict concrete CS. As shown in Fig. 18, a comparative analysis with previous research on CS prediction is presented. Liu [106] achieved an exceptionally high accuracy with an R^2 -value of 0.999 using the XGB model. However, it should be noted that this result was obtained with a relatively small dataset of 60 samples, which may not generalize well to larger datasets. Liang et al. [107] and Hongwei et al. [19] conducted similar studies using smaller datasets of 190 and 98 samples, respectively. Their best models achieved high R^2 -values (0.947 and 0.950, respectively) but had higher RMSE and MAE values. Satish et al. [108] developed an XGB model with better performance on a moderately large dataset of 633 samples, achieving an R^2 -value of 0.950, RMSE of 3.06 MPa, and MAE of 2.13 MPa.

In this research, the SM-LR model achieved an R^2 -value of 0.953, an RMSE of 3.315 MPa, and an MAE of 2.493 MPa, demonstrating exceptional performance on a large dataset of 1030 samples. Additionally, Feng et al. [109] developed an Adaptive Boosting (AdaBoost) model on the same dataset, achieving an R^2 of 0.940, RMSE of 1.93 MPa, and MAE of 1.43 MPa. Elhishi et al. [110] reported an R^2 of 0.910, RMSE of 4.37 MPa, and MAE of 3.04 MPa, indicating a lower correlation and higher error metrics compared to the SM-LR model. In summary, the SM-LR

model in this study outperformed similar models from other studies with comparable dataset sizes, particularly in terms of the R^2 -value.

5.8. Developed an interactive GUI

This section highlights significant advancements that effectively address the practical requirements of engineers and designers for the efficient utilization of machine learning models. The complexities of database compilation, model training, and testing have historically impeded the incorporation of machine learning into everyday design tasks. To overcome these challenges, an innovative approach was developed: a Python-based web application featuring a model with optimized hyperparameters accessible via a user-friendly GUI. This GUI was developed specifically for predicting the CS of concrete, as illustrated in Fig. 19. The GUI provides a streamlined layout where users can input values for various variables, with the predicted CS dynamically displayed based on these inputs. To encourage broad access and support collaborative enhancements, the GUI has been hosted on GitHub, making it available for use and further development by the community. This initiative democratizes the use of advanced predictive models and invites contributions to refine and adapt the tool to meet various specific needs within civil engineering. The GUI can be accessed at the following URL: https://github.com/mkamel24/CS_XGB.

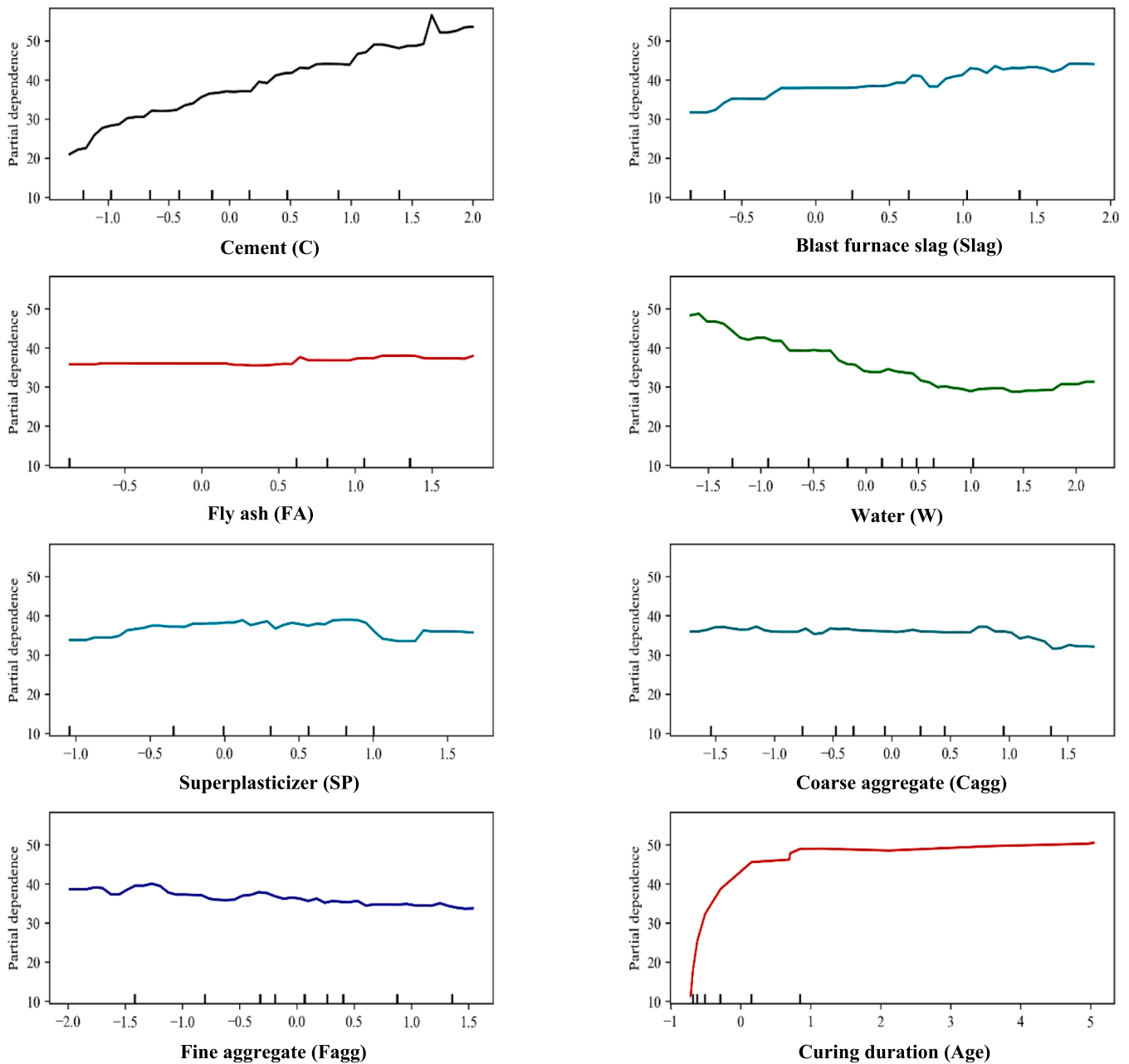


Fig. 17. PDP analysis results for the studied variables.

To assess the generalizability and reliability of the developed GUI tool for practical applications, it will validate its performance on an independent dataset, different from the one utilized in the current study for concrete CS prediction. The efficacy of the developed GUI tool will be verified by utilizing it on external experimental samples and comparing the predicted CS values with the actual experimental outcomes. To this end, it employed the dataset from the research conducted by Song et al. [60], used as a reliable external benchmark to evaluate the predictive accuracy and robustness of the developed GUI tool.

Fig. 20 presents a comparative plot of the predicted CS values (derived from the GUI tool) and the actual CS values (obtained from Song et al.). The developed GUI tool exhibits excellent predictive accuracy on the independent dataset. The findings indicate an R^2 value of 0.977 and an RMSE of 2.210 MPa, signifying a strong correlation between the predicted and actual compressive strength values, along with minimal prediction errors. This indicates that the tool offers a highly dependable predictive model with exceptional fit. Moreover, the

majority of data points for the predicted CS-values are closely aligned with the equality line and the deviation lines of $\pm 10\%$, thereby underscoring the accuracy and precision of the GUI tool. The proximity to the equality line signifies that the tool's predictions closely match the actual results, and the minor deviations within the $\pm 10\%$ range underscore its reliability and consistency. In conclusion, the performance of the GUI tool on the independent dataset confirms its superior predictive capabilities. This affirms the tool's dependability and applicability for practical use, offering it a vital tool for predicting concrete compressive strength across various scenarios.

6. Conclusions

In this research, an extensive analysis of 1030 datasets on CS was conducted, utilizing information from previous journal papers. The main aim of this study was to compare the performance of a Stacked Model based on Linear Regression (SM-LR) with base models (BM):

	< 0.880	0.880 ~ 1.864	1.864 ~ 2.848	2.848 ~ 3.832	3.832 ~ 4.816	4.816 ~ 5.800	> 5.800
Present study SM-LR	0.953	3.315	2.493				
Elhishi et al. (2023) XGBoost	0.910	4.370	3.040				
Satish et al. (2023) XGBoost	0.950	3.060	2.130				
Liang et al. (2023) BP-GA	0.947	5.800	4.490				
Liu (2022) XGBoost	0.999	1.372	--				
Hongwei et al. (2021) Bagging	0.950	4.970	3.690				
Feng et al. (2020) AdaBoost	0.940	1.930	1.430				
	R²	RMSE (MPa)	MAE (MPa)				

Fig. 18. Comparative analysis for prediction of the concrete CS.

Prediction of Concrete Compressive Strength

GUI Model for Predicting Concrete CS

Developers: Mohamed K. Elshaarawy, Abdelrahman K. Hamed & Mostafa M. Alsaadawi

Definition of Parameters

X1: Cement (kg/m³)

X2: Blast Furnace Slag (kg/m³)

X3: Fly Ash (kg/m³)

X4: Water (kg/m³)

X5: Superplasticizer (kg/m³)

X6: Coarse Aggregate (kg/m³)

X7: Fine Aggregate (kg/m³)

X8: Age (day)

Output:

Concrete compressive strength (MPa)

Fig. 19. GUI screenshot for calculating the CS-values of concrete.

ANN, RF, and XGB, including eight input parameters. The dataset is divided into three separate sets: training, validation, and testing. Both training and validation datasets were used to develop and test the BMs. While the SM-LR was built using the developed BMs predictions in the validation stage, then the last testing dataset was used to evaluate its performance. Comparative analysis was conducted to assess the predictive accuracy of both types of models. The study yielded the following main findings:

1. Among the base models, the XGB model demonstrated superior predictive accuracy during the testing stage, with an R^2 value of 0.947 and an RMSE of 3.524 MPa.
2. The SM-LR model outperformed all base models, achieving the highest R^2 of 0.953 and the lowest RMSE of 3.315 MPa, which underscores its superior predictive capabilities.
3. SHAP and PDP analyses indicated that the most critical factor was the age of the concrete, which consistently improved predictions as it increased. The second most important input was the cement content, which had a threshold effect that significantly influenced predictions up to a certain level. Conversely, factors such as blast-furnace slag and coarse aggregates exhibited moderate effects, whereas fly ash, superplasticizer, and fine aggregates had the least influence.
4. Rank analysis showed that among the BMs, the XGB stood out, followed by the RF model, which demonstrated good accuracy but had slightly higher errors. The ANN models ranked lowest, with lower correlation and higher errors.
5. A user-friendly GUI was developed to make the predictive models accessible for real-world applications. The GUI, available on GitHub, allows engineers to input concrete mix parameters and receive

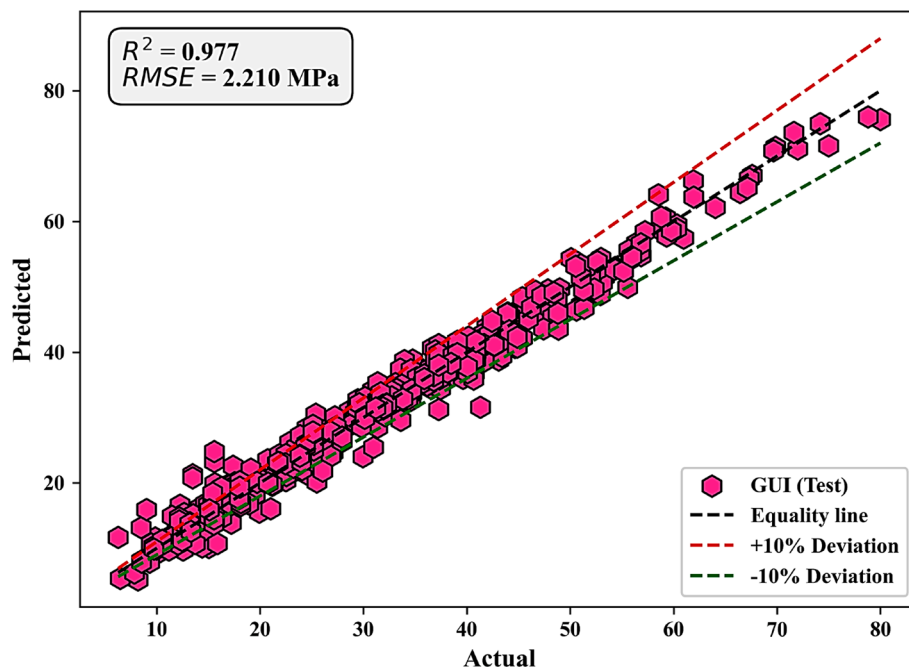


Fig. 20. Validation of developed GUI tool against actual experimental results from Song et al. [60].

predicted CS values in real-time, enhancing the usability of these models in practical design scenarios.

7. Limitations and future work

While the developed model demonstrates strong predictive performance, several limitations should be noted. The model's effectiveness heavily relies on the quality and representativeness of the input data. Since the dataset used in this study does not include key factors such as temperature, humidity, and curing methods, its ability to generalize to other environments or experimental setups may be limited. Furthermore, the model does not account for all variables that may influence concrete CS, such as specific mix proportions and external environmental conditions, potentially affecting its accuracy in certain real-world scenarios. Additionally, while the GUI is designed for simplicity, users must still have an understanding of the input parameters and their relevance to make the most effective use of the tool.

Future work should focus on expanding the dataset to incorporate a broader range of concrete compositions, environmental conditions, and laboratory variables, including those that might affect the curing process. Moreover, integrating advanced deep learning models could enhance predictive accuracy by identifying more complex relationships in the data. The GUI could also be improved by adding interactive features, guidance, and tutorials to help users with varying levels of expertise navigate the input process. Finally, validating the model using real-world data from diverse applications and environmental conditions will be crucial to ensure its robustness, generalizability, and practical applicability.

CRediT authorship contribution statement

Abdelrahman Kamal Hamed: Funding acquisition, Formal analysis, Data curation, Conceptualization. **Mohamed Kamel Elshaarawy:** Software, Resources, Project administration, Methodology, Investigation. **Mostafa M. Alsaadawi:** Writing – review & editing, Writing – original draft, Visualization, Validation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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